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LETTER FROM THE EDITOR

Dear Reader,

Since 2021, Anthropy has sought to cultivate intellectual curiosity and meaningful engagement within the humanities and social sciences.

Through initiatives such as Humanities Focus Week and our flagship In2Academia research programme, we provide students with opportunities to engage rigorously with academic ideas, learn from experts across disciplines, and pursue independent research under dedicated mentorship.

It is therefore with great pride that we present the third edition of the Anthropy Review. This journal showcases the work of budding student researchers who have spent months developing original questions, engaging with academic literature, and refining their arguments in their first time through the research process.

The articles that follow reflect not only the breadth of interests within our community, but also the intellectual ambition, curiosity, and diligence that underpin scholarly inquiry.

We hope this edition offers both an engaging read and a glimpse into the exciting research being undertaken by the next generation of scholars across disciplines in the Humanities.

*Sincerely,
Ruhani Surana
Editor-In-Chief,
Anthropy Review*



ABSTRACT

This issue of The Anthropy Review presents a diverse range of research, each paper offering a fresh perspective on complex topics related to identity, culture, and power. Together, they explore a variety of ideas and frameworks that challenge traditional boundaries.

To begin, Anahita Ghosh Singh (UWCSEA), in “The Abortion Act 1967 and the Illusion of Reproductive Autonomy in British Law”, delves into how the 1967 Act acted as a paternalistic compromise that restricted bodily autonomy through medical gatekeeping.

Next, Derek Shih (UWCSEA) examines the dramatic shift in EU trade relations and integration in “Testing European Unity: How U.S. Tariffs Reshaped Economic Interdependence in the EU”, demonstrating how U.S. tariff shocks forced the bloc toward strategic autonomy and deeper economic interdependence despite internal political constraints.

Leyan Ji (UWCSEA), meanwhile, in “From Politics to Economic: Evaluating the Origin of Modern Japanese Nationalism” highlights the pivotal role of economic dissatisfaction and stagnation in driving right-wing nationalism in the Japanese population.

Shifting focus to geography, Katherine Wang (UWC Changshu China) explores the impact of crop straw burning on air quality and resident wellbeing in “To what extent does the reduction of crop straw burning contribute to improving air quality and the resulting resident health in Baguazhou, Nanjing, China, from 2010 to 2025?”.

Yueqi Gou (UWCSEA)’s “Take Your Meds: Reducing Motivational Barriers for Non-Adherence Hypertensive Patients Through Incremental Mindsets” provides an intriguing analysis of how intrinsic motivation and growth mindsets interact with patient nonadherence.

This journal represents a three-month process of dedication and collaboration, with ten sessions devoted to refining each paper through feedback and discussion. We would like to thank all the mentors who devoted their energy and time to guiding the students throughout this process. Their support has been invaluable in shaping these papers and ensuring the quality of this journal.



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THE ABORTION ACT 1967 AND THE ILLUSION OF REPRODUCTIVE AUTONOMY IN BRITISH LAW

WRITTEN BY ANAHITA GHOSH SINGH



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ABSTRACT

This paper challenges the traditional view of the UK's Abortion Act 1967 as a definitive victory for women's rights, arguing instead that it functioned as a paternalistic political compromise. By analyzing the legislation through political, legal, and social lenses, the essay demonstrates how the Act shifted reproductive control from the courtroom to the clinic rather than granting genuine bodily autonomy. It examines the "two-doctor rule" and statutory frameworks as systems of medical gatekeeping that infantilized patients, medicalized socio-economic struggles, and created severe regional inequalities. Far from being a final destination for liberation, the paper concludes that the Act's systemic limitations and conditional access stripped women of self-determination, ultimately serving as the foundational grievance that catalyzed the radicalism of the UK's Women's Liberation Movement.

INTRODUCTION AND LITERATURE REVIEW

The 21st century has been defined by a significant shift in the legal and social frameworks governing reproductive health. To understand this evolution, it is necessary to examine the specific catalysts that have altered access and regulation over the last several decades. In 1967, the British state stopped treating women as criminals for seeking abortions, but it did not yet trust them to be the masters of their own bodies. While the Abortion Act 1967 is frequently considered a landmark victory that ended the era of "backstreet" procedures and significantly reduced maternal mortality, its legacy as a tool for empowerment remains deeply contested. This paper will argue that the Act did not significantly advance reproductive rights in the UK as it functioned more as a political compromise than a true expansion of women's autonomy. Although on paper women were given fair access, due to geographical, monetary, and religious factors, true access was not available to a significant portion of the nation. Doctors were designated as legal gatekeepers, meaning that women required a medical justification for personal choices; however the doctor's morals and beliefs did not guarantee true, unbiased access. This role is essential for maintaining standards, but it can also restrict access. The gatekeeper role may be essential for maintaining standards across fields, it can also restrict access to needs an individual may have. The legislation merely shifted control from courts to the clinic. Ultimately, while the Act advanced public safety and established abortion as a legitimate form of healthcare, it failed to provide the dignity of self-determination. To evaluate the scope of the 1967 Abortion Act, this paper examines its political, legal, and social impacts through three interconnected lenses. The extent to which the Act advanced reproductive rights is assessed by tracking the persistence of medical gatekeeping via regional statistical disparities in healthcare access; analyzing the tension between legal decriminalization and genuine autonomy through the medicalization of socio-economic struggles; and evaluating the law's role in catalyzing the radicalization of the Women's Liberation Movement through the linguistic transition from welfare-based reform to radical grassroots activism. Academic discourse around the Act is characterized by a tension between its status as a "humane and far-sighted" social reform and its function as a paternalistic tool of medical control. Early historical assessments often align with the perspective found in the ICBH Witness Seminar, which reconstructs the Act's passage through the testimony of the political and medical elites who crafted it. This testimony highlights that the act was born out of necessity; the pre-1967 legal landscape, dominated by the Offences Against the Person Act 1861 and the Preservation Act 1929, had created a crisis of unsafe illegal abortions that the narrow medical defense offered by *R v Bourne* 1938 could not resolve. However, the ICBH account also exposes the Act's primary weakness: it was a product of intense political compromise. To make the bill pass and get supported by the British Medical Association (BMA) and the Abortion Law Reform Association (ALRA), the goal of total autonomy was abandoned, allowing medical professionals to hold the

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ultimate power over reproductive care. Through the lens of feminist theory, the implications of this compromise are further scrutinized. As noted in David Publishing's analysis, the Act's "neutral" rules, such as Section 1(1)(d) regarding fetal abnormality and the Conscientious Objection clause, reflect a legal framework that prioritizes professional ethics and state interest over the principle of "My Body, My Choice." (2) This perspective suggests that the act did not actually grant a right to abortion but rather created a legal exception to criminal prosecution. While some biographers of the law describe it as a pinnacle of 20th-century legislation, feminist theorists argue that it bypassed the core requirement for reproductive autonomy.

Ultimately, the existing literature reveals significant gaps that this essay seeks to address. Existing literature often fails to integrate political, legal and social dimensions necessary to prove this law's ineffectiveness. Although critique is typically focused on the medical framework, which aligns with what I attempt to prove, existing literature often fails to address how the legislation functions as an assault on women's dignity. Additionally, existing research often details the negotiations of medical elites and the theoretical shortcomings of the medical model, there is typically less focus on how these legal compromises specifically fueled the radicalism of the Women's Liberation movement and future legislation. By examining the Act not as a final destination but as a catalyst that politicized women through its own limitations, this essay will bridge the gap between the Act's clinical success and its failure to secure genuine reproductive rights. The Abortion Act 1967 did not significantly advance the reproductive rights movement in the UK, as it functioned more as a political compromise than a true expansion of women's autonomy, placing decision-making power in the hands of doctors rather than women themselves and thereby limiting genuine reproductive freedom.

SECTION 1: POLITICALLY

THE 1861 ACT

The Offences Against the Person Act 1861 was a legal framework that consolidated all violent crimes into a simple statute to simplify the law. Although now much of it has been updated, it still serves as a strong foundation in British Law. Although this act held its benefits, under Sections 58 and 59 of the Act, abortion was considered a criminal offense. Access to it was banned, with no exceptions, providing considerable harm to women. In most cases, if a woman were even at risk, it was the law that the birth must occur, regardless of the consequences it may bear. By the mid-20th century, the

1861 Act was no longer viewed as an effective deterrent but as a direct cause of a public health crisis, characterized by thousands of backstreet abortions that left women dead or permanently mutilated. Data from the Royal College of Obstetricians and Gynaecologists (RCOG) prior to the 1967 Act estimated that between 50,000 and 100,000 illegal abortions were performed annually in the UK, resulting in an average of 40 to 50 preventable maternal deaths each year and accounting for roughly 14% of all maternal mortality. The 1861 acts had resulted in significant harm that required immediate action; however, that action was rushed and forced rather than well-thought-out. The 1967 Act was passed primarily out of a sense of urgency, operating under the assumption that any legislative intervention would be preferable to the status quo. Parliament was driven by stark data showing that up to 100,000 dangerous illegal procedures were occurring annually in the UK. This pragmatic urgency was echoed by the bill's sponsor, David Steel MP, who argued that immediate reform was vital because the existing law had completely 'broken down' and was 'neither respected nor obeyed. Although not intentional, this large public health crisis created by the 1861 act simply made rectification simpler. Rather than offering the best option, the state simply provided something slightly better, convincing individuals that the problem was fixed, though it was far from it.

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THE 1929 ACT

The Infant Life (Preservation) Act simply added to the turmoils that women faced. A large gap that surfaced in the 1861 act was that there was no law in order to prevent the murder of a child while being born. This act bridged that gap, further lessening the reproductive rights that women held. However, this act's inclusion of a narrow exception for "preserving the life of the mother" inadvertently created a legal gray area. This was famously exploited in the 1938 R v Bourne case, where the court ruled that a doctor could perform an abortion if the birth were to "physically or mentally wreck" the woman. Justice Macnaghten's ruling in R v Bourne (1938) wasn't meant to be a conspiracy to disempower women; it was a tactical, defense-led response to a horrific crime (the gang rape of a 14-year-old girl) designed to protect a well-meaning doctor from a life sentence under the 1861 Act. Although one can argue that while Bourne's intent was a narrow, humanitarian rescue of a specific physician, its structural legacy was the institutionalization of the doctor as the ultimate moral arbiter. This precedent simply reinforced the "medical model" of abortion as it established that a woman's needs were only valid when filtered through the subjective lens of a physician's diagnoses. The 1929 Act acted as a bridge to the 1967 compromise by proving that the state was willing to provide abortions as if it were only a medical concession. Thus, the 1967 act didn't invent reproductive rights; it simply codified the medical gatekeeping that had been simmering since the 1920s, ensuring that the doctor, not the patient, remained the primary legal actor.

THE JUSTIFICATION

The combined legacy of the 1861 and 1929 acts served as a powerful historical justification for the 1967 reform, allowing proponents to frame the new law as a compassionate and moral necessity. By highlighting the horrors of the "backstreet" era, reformers could position the new Act as a national apology for the century of state-enforced trauma. However, this framing was a double-edged sword. By focusing the justification on public health and the prevention of harm rather than bodily autonomy, the political architects of the bill avoided the radical implications of true reproductive freedom. The apology was fundamentally conditional; it admitted the harshness of the previous laws, but it did not admit the lack of choice women faced. Instead, it framed women as the victims of their own circumstances who required the protection of a legalized medical framework. This placed women in a powerless situation, as they were treated as incapable individuals who needed choices made for them. Although the new Act did legalize abortion, access to it was not at the woman's choice, highlighting the injustices within the political system. The act rebranded control as care, effectively highlighting the unjust power dynamic of society, where the woman's life is still in the hands of others.

THE COMPROMISE

Ultimately, the 1967 Abortion Act was an exercise in pragmatic political accommodation that prioritised legislative consensus over the advancement of reproductive rights. Introduced by Liberal MP Sir David Steel and facilitated by the Labour government, the act was ingeniously engineered as a compromise to pacify conservative and religious opposition. In 1967, the Conservative Party was in opposition, and according to polling data from that period, they were highly popular, with support often exceeding 50%. If the Act were to be too "liberal," it would have faced immense opposition, leading to protests throughout the country. To avoid this, Sir Steel chose to make a compromise. He reintroduced the concept of medical professionals being the judges. According to socio-legal scholars Sally Sheldon and Carol Smart, this structural choice meant that the 1967 Act did not secure an inherent reproductive right, but instead positioned abortion as a clinical concession granted to women by the medical establishment.

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The act mandated that 2 Doctors would have to provide medical reasoning and certification in order for the woman to receive it. While proponents argued that the two-doctor requirement served as a necessary rational safeguard to prevent unilateral medical malpractice and minimize clinical harm, in practice, this mechanism functioned less as a protective measure and more as an institutional barrier that delayed time-sensitive healthcare. The conscientious objection clause added to abortion illustrates that the state prioritised the moral comfort of a doctor over the genuine needs of a woman. Consequently, while the 1967 act succeeded in making abortion safe and legal, it was unable to advance reproductive rights significantly and introduced a system of paternalistic gatekeeping.

SECTION 2: LEGALLY

THE STATUTORY DISPLACEMENT OF PATIENT AUTONOMY

The 1967 Abortion Act is often misunderstood as a document that granted women a right to choose; however, a strict legal analysis reveals it functioned as a statutory defense for medical professionals. This distinction is vital because it establishes that the law did not grant positive liberty to women. Instead, it created a system which scholars call a “gatekeeper system” where the legal focus of control was shifted from the courtroom to the consultation room. As seen in *Royal College of Nursing v DHSS* [1981], the House of Lords confirmed that the doctor’s involvement was the essential legal requirement for the procedure to remain lawful. Consequently, the patient’s will was considered legally inferior to the doctor’s clinical authorization. This reinforces a hierarchy where medical permission outweighs personal choice and comfort. The difference between keeping and terminating the fetus would have dire consequences for the mother; the law oversaw this and chose to provide the difficulty of the abortion decision to doctors. Overall, this act was a method to keep courtrooms calmer, serving as a simple justification rather than true justice.

LEGAL INFANTILIZATION THROUGH THE "TWO-DOCTOR" REQUIREMENT

The requirement that two separate medical practitioners certify an abortion created a unique procedural hurdle that effectively infantilized the pregnant individual. This “two-doctor rule” is a prime example of legal paternalism, a concept where the state restricts a person’s liberty for their “own good.” Legally, this mandate codified the suspicion that a woman is an unreliable arbiter of her own moral and physical necessity. It mirrors the logic of earlier case law, such as *R v Bourne*. By formalizing this requirement, the 1967 Act suggested that a woman’s testimony about her own life was legally insufficient. She was not treated as a full legal actor but as a subject of care who required a peer-reviewed consensus to validate her private decisions. An individual has stronger insight into their own lives and behaviors, but the court undermined this completely, placing all decision-making capacity in the hands of two strangers. The moral and subjective views of the doctor were to be heard, while the woman would be infantilized. This highlights the lack of advancement in reproductive rights that this act brought about.

JUDICIAL DEFERENCE AND THE GOOD FAITH STANDARD

When evaluating the extent to which the 1967 Abortion Act advanced reproductive rights, the “Good Faith” standard emerged as a profound paradox that functioned as a double-edged sword for women. On one hand, it was not a practical barrier to access; by shielding clinical decisions made “in good faith,” the Act allowed sympathetic physicians and charitable providers to interpret the law broadly. This created a vital mechanism through which safe, widespread abortion services could be established under the guise of medical necessity. On the other hand, the standard acted as a distinct conceptual barrier to true bodily autonomy. This legal tension was solidified in *Paton v British Pregnancy Advisory Service* [1979], where the High Court heavily deferred to medical authority, ruling that because the statutory requirements had been met by certifying doctors, the judiciary would not intervene.

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While Paton successfully protected the scheduled procedure from external interference, its underlying rationale reinforced that the legal right to decide belonged entirely to the medical profession rather than the patient. Consequently, the 1967 framework advanced reproductive rights only to the extent of medicalization; it secured practical access through a loophole of professional discretion, but simultaneously institutionalised a paternalistic system where a woman's personal liberty remained legally contingent upon obtaining mandatory doctor approval.

THE MEDICALIZATION OF SOCIO-LEGAL RIGHTS

By framing the grounds for abortion around “mental or physical health,” the Act forced social and economic issues into a medical framework. This is a process known as medicalization, where non-medical problems are treated as medical disorders. Legally, a woman could not seek an abortion simply because she was in poverty; she had to present her poverty as a threat to her mental stability. This shows how the act didn't advance rights, as a woman who understood that she did not have the capacity to provide for the child was forced into the situation due to a lack of fair representation. She could not appeal or sue, as the act clearly stated that the doctor's words were final. This requirement forced women to perform a state of distress or pathology to satisfy the law. This subordinates a person's legal agency to a medical diagnosis, reinforcing the paternalistic idea that a woman's choices are only valid when they are translated into a clinical crisis. Legal consistency was prioritized over individual autonomy to turn its back on the true implications of the child's birth, and simply focused on the prevention of “murder.”

SECTION 3: SOCIALLY

SOCIAL INEQUALITY IN MEDICAL ACCESS

The 1967 Act brought about a system that was quite similar to a postcard lottery. Because the law relied on the total discretion of local doctors and the National Health Service (NHS) was managed by regional boards, access to an abortion became entirely dependent on where a woman resided. In areas with conservative or religious views, such as Birmingham and Liverpool, doctors often used their power to push their own moral beliefs and block any requests that opposed them. Conversely, in more liberal areas, it became far easier for women, creating a healthcare disparity throughout the nation. This geographical divide created a social hierarchy; wealthier women could travel to cities such as London in order to gain access to clinics that would bypass restrictive local doctors. Working class women were bound by the moral views of their local gatekeepers as they would not be able to afford travel to larger cities. This suggests that the law favoured the privileged, allowing them to have an easier route to terminate the pregnancy compared to those of the working class. Although each woman holds their own struggles, many working-class women strongly require abortion, as long maternity leaves, often lowly paid, would leave them in a predicament that could harm them and their child. In a low-income family, one child could result in the entire family going hungry. In addition, the birth of children in low-income families results in a higher volume of orphaned children. Based on a July 2022 report by the Parliament's Joint Committee on Human Rights (JCHR), it is estimated that 185,000 babies were born to unmarried mothers and subsequently adopted in England and Wales between 1949 and 1976. One would assume that post the 1967 Act, many of the nationwide issues regarding childcare would be solved, it can be shown that this is far from the truth.

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SACRIFICING DIGNITY FOR CARE

From a public healthcare perspective, the Act was a triumph; statistics show that the number of recorded deaths due to abortion in England and Wales dropped from 160 during 1961-63 to just 9 during 1982-84. There was a massive decline in emergency room admissions for “septic” backstreet abortions. However, this safety came at the cost of social dignity. To satisfy the medical grounds of the Act, women were often forced to perform a state of mental instability or extreme distress. They had to prove they were broken or unfit to deserve the procedure.. This turned the social interaction between a doctor and a woman into a power struggle. If a woman appeared too calm, certain or rational, she risked failing the criteria by looking too stable to qualify for medical intervention, creating a structural paradox where she had to prove she was mentally broken to get care, yet rational enough to provide informed consent. Instead of a patient making a rational life choice, the woman became a supplicant, which reinforced the social stigma that seeking an abortion was a sign of failure rather than an exercise of autonomy. This bureaucratic trauma meant women had to rehearse this psychological crisis for clinical gatekeepers who held the subjective power to validate or dismiss their sanity. This proves the weakness of the act as it did not provide a proper objective framework, but rather one that could be open to interpretation. Ultimately, by forcing healthy individuals to pathologize their lives to secure basic health care, the legal framework systematically removed female agency and turned what should have been a right into a privilege.

MEDICAL PATERNALISM VS. PERSONAL AGENCY:

The Act moved the control of women’s bodies from the illegal backstreet into the legal clinic, but it did not grant the control to women themselves. By making the doctor the final judge, the state reinforced a paternalistic social structure where professional (and historically male) authority overrode female agency. Socially, the Act encouraged a culture where women were expected to be grateful for a doctor’s concession. This prevented abortion from being viewed as a standard part of reproductive healthcare and instead kept it framed as a moral favor granted by the medical establishment. This dynamic maintained traditional gender roles, as women were kept in a subordinate position where an external authority figure legally signed off on their economic and professional futures. Sociologically, this mechanism did not simply regulate a clinic procedure; it institutionalised a system of clinical dominance, granting the medical establishment unprecedented power to define, police, and validate normal female behaviour. By forcing women to bargain for clinical approval rather than claiming an inherent statutory right, the legislation effectively modernized the apparatus of reproductive surveillance rather than dismantling it.

THE RISE OF THE LIBERATION MOVEMENT

Many historians argue that the 1967 Act was a “failed promise” that actually helped grow the Women’s Liberation Movement (WLM). The frustrations of dealing with bureaucratic delays and judgmental doctors politicized an entire generation of women. Activists targeted the paternalism of the two-doctor rule, which was a major legal act that stripped women of moral and bodily agency and fiercely rejected the criteria that forced women to pathologize themselves as mentally fragile. These explicit legal barriers led directly to the rise of radical grassroots organizations like the National Abortion Campaign (NAC) in 1975. Their slogan, “Abortion on Demand – A Woman’s Right to Choose,” was a direct social reaction against the limitation of the 1967 Act. Ultimately, the social impact of the act was to prove that the structural limits of the Act demonstrated that state-granted tolerance is fundamentally distinct from genuine freedom. Under the guise of a progressive reform, abortion remained a conditional medical privilege rather than a realized right to self-determination.

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Although the act did not manage to advance reproductive rights significantly, the social reactions to it brought about serious change in the UK. By turning clinical restrictions into political fuel, the WLM effectively transformed the Act from a flawed piece of legislation into the foundational grievance of modern British feminism.

CONCLUSION

Ultimately, the 1967 Abortion Act has been fundamentally mischaracterized as a landmark victory for reproductive rights; in reality, it traded the radical demand for bodily autonomy for a highly regulated public health concession, cloaking new legal restrictions and clinical gatekeeping in the rhetoric of liberation. It did succeed in preventing an era of backstreet abortions, limiting maternal mortality, which is something the act truly should be commended for; however, fair access to abortions was still incredibly limited, thus making it a paternalistic law. Politically, the Act reflected caution rather than conviction. It was shaped by compromise, designed for an adequate solution rather than a transformative one. It was justified solely on public health grounds rather than on individual rights. Legally, this compromise was embedded in its structure, where abortion existed not as a right but as a conditional exception, dependent on the approval of doctors. This requirement of medical validation and the protection of professionalism and discretion reinforced a hierarchy that placed women's loves and experiences beneath clinical judgment. Socially, These limitations were even more visible. Access varied depending on geography and class, resulting in a disparity of abortion access for women in the UK. While the act did offer care, it often did so at the expense of dignity, reinforcing a system where women had to seek permission rather than exercise choice. However, the UK's legislative landscape has since fractured, creating an uneven landscape of reproductive rights. While the 1967 Act still governs England, Scotland, and Wales, Northern Ireland fully decriminalized abortion in 2019, and Scotland later pioneered the buffer-zone format. This statutory divergence carries a profound implication: it strips the UK of a unified standard for bodily autonomy, reducing reproductive healthcare access to a "postcode lottery." Ultimately, this uneven progress proves that the 1967 Act remains a stagnant, centralized framework that has failed to guarantee equal rights across the entire nation. Although under the Crime and Policing Act 2026, abortion has become fully decriminalised for women in England and Wales, the regulatory two-doctor rule remains a requirement for medical providers to ensure that the procedure is performed within the legal framework. Ultimately, the 1967 Act was a public health compromise rather than a radical revolution for women's liberation. By replacing criminal penalties with the "two-doctor rule," it established a framework of medical gatekeeping rather than absolute bodily autonomy. Today, the British Pregnancy Advisory Service's (BPAS) frustration with bureaucratic delays and the British Medical Association's (BMA) push for full decriminalization highlight a critical conceptual takeaway: legislative legalization does not guarantee equitable access. The ongoing struggle for reform proves the UK is still transitioning away from this paternalistic framework toward genuine healthcare equity. While the struggle for women's reproductive autonomy began in the nineteenth century, modern equality remains unrealized as systemic barriers continue to restrict actual access to essential healthcare services.

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TESTING EUROPEAN UNITY: HOW U.S. TARIFFS RESHAPED ECONOMIC INTERDEPENDENCE IN THE EU

WRITTEN BY DEREK SHIH

TESTING EUROPEAN UNITY: HOW U.S. TARIFFS RESHAPED ECONOMIC INTERDEPENDENCE IN THE EU

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ABSTRACT

This paper examines how the weakening of Pax Americana through U.S. reciprocal tariffs (2025–2026) has affected economic interdependence within the European Union (EU). Using a qualitative, document-based methodology, it analyses policy reports, economic studies, institutional publications, and trade data from sources including the European Parliament, European Commission, IMF, Reuters, and Bruegel. The findings show that U.S. tariffs disrupted transatlantic trade by reducing EU exports, increasing investment uncertainty, and weakening confidence in the American-led economic order. In response, the EU accelerated efforts to strengthen strategic autonomy, improve supply-chain resilience, deepen defence cooperation, and enhance capital-market integration. However, deeper collaboration remains constrained by differing national interests, uneven exposure to tariffs, and complex decision-making processes. While full economic detachment from the United States remains unlikely, the tariff shock has encouraged greater European coordination and self-reliance. Overall, the study concludes that the decline of Pax Americana has strengthened EU economic interdependence, though integration remains limited by political and institutional challenges.

RESEARCH QUESTION

To what extent does the fall of pax America due to reciprocal tariffs from the United States (2025-2026) affect the economic interdependence of the European Union?

INTRODUCTION

The idea of Pax Americana refers to the post-1945 international order in which the United States played the central role in securing global stability through trade leadership, military protection, and the spread of liberal economic rules such as reducing trade tariffs and establishing the USD to be the world's primary reserve currency. During this time period, US hegemony catalysed international stability through the formation of multi-lateral institutions like the IMF and the UN as well as security alliances (NATO) that was contingent on the American dominance in military power (International Monetary Fund, Oct. 2025) . That order did not mean the absence of conflict, but it did mean that many allies, including European states, could rely on American power to sustain predictable trade and security conditions. In 2025 and 2026, however, reciprocal tariffs and a more volatile U.S. trade posture weakened the transatlantic environment, forcing the European Union to reassess the extent to which its prosperity could continue to depend on America's willingness to uphold open markets. This development emerged from a broader shift in U.S. trade policy, in which tariffs were increasingly deployed as instruments of economic pressure rather than as exceptions within a rules-based order. As a result, European policymakers began to question the reliability of the United States as the central guarantor of global trade stability. The EU's response has therefore moved beyond short-term concern over exports and toward deeper questions of strategic autonomy, supply-chain resilience, and the long-term security of European economic interdependence. (European Parliament Research Service, 2026).

This paper will explore the deconstruction of Pax Americana due to reciprocal tariffs from the US and whether or not it has accelerated economic cooperation inside the EU, especially through strategic autonomy, supply-chain resilience, and joint investment.

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METHODOLOGY

This paper adopts a qualitative, document-based methodology to examine how the weakening of Pax Americana through U.S. reciprocal tariffs in 2025–2026 affects the European Union’s economic interdependence.

The methodology is grounded in policy analysis and comparative interpretation of secondary sources. It draws on European Parliament briefings and studies, European Commission Security Action for Europe material, IMF regional economic reporting, Bruegel and Center of Economic Policy Research analysis, Intereconomics commentary, and websites such as Reuters and ING reporting on tariff effects. These sources are used because they combine institutional evidence, macroeconomic context, and current trade data, allowing the paper to connect short-term tariff disruption with broader structural changes in the EU economy.

A major gap in the current literature is that U.S. tariffs are often discussed mainly as a transatlantic trade issue or as a temporary export shock. Less attention is given to how tariff volatility changes the internal dynamics of the EU itself. This paper addresses that gap by showing that external trade pressure can increase intra-EU interdependence, making cooperation on investment, resilience, defence, and supply chains more necessary. In this sense, the study shifts the focus from the tariff shock alone to the way European states respond collectively to it.

The data used in this paper are secondary and mostly qualitative, supplemented by selected quantitative indicators. These include export declines, trade-surplus changes, and growth forecasts that illustrate the economic consequences of tariff instability. The evidence also covers policy developments in defence funding, energy security, and capital-market integration, which help demonstrate how the EU is adapting institutionally. Together, these materials support a broader political-economic reading of the tariff shock.

This method is appropriate because the research question is structural and interpretive rather than purely numerical. A qualitative case-study approach makes it possible to trace how external pressure influences European decision-making across several policy areas at once. It is also suited to a recent and still developing issue, where institutional responses and economic effects are evolving too quickly for a narrowly statistical approach to capture the full picture.

LITERATURE REVIEW

The existing literature on multipolarity, strategic autonomy, and the changing international order provides important theoretical foundations for understanding how the weakening of Pax Americana may influence economic interdependence within the European Union (EU). While scholars generally agree that the international system is becoming more multipolar, they differ on the extent to which the EU can emerge as an autonomous geopolitical actor and on the mechanisms through which it can respond to declining American leadership.

Bernd von Muenchow-Pohl argues that the twenty-first century is characterised by a redistribution of power away from the traditional Western core and towards a more multipolar international order. He suggests that globalization, shifting economic power, and the growing importance of non-traditional security challenges require actors such as the EU to redefine their role in global governance. The author questions whether the EU can become a genuine strategic actor capable of projecting influence beyond its immediate neighbourhood and contributing to the management of global order. This perspective is valuable because it frames the EU not simply as an economic bloc

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but as a potential geopolitical power operating in a post-American international system.

Similarly, Okano-Heijmans and Sundar argue that growing uncertainty generated by Brexit, U.S. trade protectionism, and China's expanding influence has increased the importance of strategic economic cooperation. They contend that geo-economics is becoming a dominant feature of international relations and that states must increasingly integrate economic and political objectives. Their analysis emphasizes the importance of sustainable connectivity, institutional cooperation, and coordinated economic frameworks as tools for maintaining prosperity and stability in an increasingly competitive international environment. The authors further argue that cooperation among like-minded partners is necessary to preserve a rules-based international order and respond to challenges posed by changing global power dynamics.

More recent literature focuses specifically on the impact of Donald Trump's return to office and its implications for Europe. Simonet argues that the second Trump administration has exposed long-standing structural weaknesses within the EU while simultaneously accelerating efforts toward strategic autonomy. According to Simonet, American hostility towards European institutions, uncertainty regarding security guarantees, and increasing economic tensions have forced European policymakers to reconsider their dependence on the United States. The author suggests that the changing transatlantic relationship acts as a catalyst for greater European cooperation, particularly in defence, industrial policy, and economic governance. However, he also notes that internal divisions between member states remain a significant obstacle to deeper integration.

Although these studies provide valuable insights into multipolarity, strategic autonomy, and the evolving transatlantic relationship, relatively little research has examined how recent U.S. reciprocal tariffs specifically affect economic interdependence within the EU. Much of the literature focuses on geopolitical competition, security cooperation, or the EU's external partnerships rather than the internal economic consequences of declining American leadership. This research seeks to address that gap by investigating whether the weakening of Pax Americana through U.S. tariff policies has encouraged deeper economic coordination and integration among EU member states. By focusing on the relationship between external economic pressure and internal European cooperation, this study contributes to ongoing debates about the future of European integration in an increasingly multipolar world.

BACKGROUND INFORMATION ON THE EU

The EU's creation was shaped by a desire to prevent another destructive cycle of war in Europe. At its core, the Union was created to promote peace, stability, solidarity, and shared prosperity through economic integration rather than rivalry. The founding logic was that if states became more economically interdependent, conflict would become less attractive and cooperation more rewarding. Over time, those ideals expanded into a broader political project built around democracy, the rule of law, equality between member states, and the protection of common European values which looks like democracy and human dignity to freedom. This is why the EU is not just an economic based partnership but also a rock solid alliance which tries to bind countries together through shared long-term commitments and goals.

The European Union is a highly institutionalised political and economic union made up of 27 member states, established on November 1, 1993. The current structure of the EU is made up of five key institutions: the European

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Commission, which has the right to propose legislation; the Council of the European Union, where national ministers negotiate and adopt laws; the European Parliament, which holds authority over legislation and budget allocation; the European Council, which dictates the political agenda and direction of the EU; and the Court of Justice of the European Union, which helps ensure EU law is applied consistently across all member states. By establishing checks and balances between the EU's competences, this methodical division ensures that there isn't a consolidation of power that allows one party to be the unipolar force in finalising all decisions.

One of the EU's most important structural features is the single market. The single market is designed to guarantee the free movement of goods, services, capital, and persons across member states, which reduces internal barriers such as developmental gaps between nations and unequal distribution of economic benefit where less developed member states struggle to compete with wealthier partners. This in turn encourages deeper economic integration by allowing all countries to expand their market size, ensure supply chain resilience, and increase foreign direct investment. This is one of the main reasons the EU is able to produce strong economic interdependence: firms can source inputs across borders, investors can move capital more easily, and workers can live and work in different member states. However, the same openness also means that economic shocks spread quickly from one country to another, making coordination necessary but also more urgent. The need for coordination is not unprecedented. During the COVID-19 pandemic, disruptions to European supply chains and shortages of critical goods quickly spread across member states, leading the EU to coordinate vaccine procurement and launch the NextGenerationEU recovery fund. This experience reinforced the importance of collective responses to cross-border economic shocks and helps explain why recent U.S. tariff measures have renewed calls for greater European strategic autonomy and internal cooperation.

The European Union is also characterised by significant economic and political differences between member states. Variations in economic size, industrial capacity, geopolitical priorities, and exposure to international trade mean that member states often enter negotiations with different interests and levels of vulnerability. These differences can complicate cooperation and make collective decision-making more challenging.

The institutional balance inside the EU creates both strength and friction. The Commission has the right of initiative, which means it is usually the institution that begins the legislative process, but it cannot impose policy alone. For most laws and trade-related measures, the Commission must work through the Council and the European Parliament, both of which can amend, delay, or reject proposals. In trade policy, the Council often uses qualified majority voting, which should in theory make decisions easier than unanimity, but in practice governments often seek consensus because trade deals are politically sensitive and affect domestic industries differently. This is especially true when agreements involve services, investment protection, or other mixed-competence questions that touch on both EU and national authority. As a result, even though the EU has the tools to act collectively, its decision-making process often requires prolonged negotiation among actors with different priorities, which slows down economic cooperation.

The institutional setup also creates a deeper problem because member states display differing perspectives and interests. Larger states such as Germany and France often have more influence because of their economic scale, diplomatic weight, and ability to shape debate through their central role in the Union. Smaller states may have less structural influence, but they can still become important in negotiations because EU rules give them formal equality

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in key institutions and because coalitions of smaller countries can sometimes block or reshape proposals. This looks like the "One State, One Commissioner" rule that ensures every country, regardless of its size, has a formally equal seat and portfolio representation on the EU's executive body. This makes the EU powerful because it can aggregate the interests of many states into a single market and a common external trade policy. However, this equally makes the EU fragile because those same states often disagree about how much authority should be centralized, how costs should be shared, and how far integration should go. This is essential towards understanding how the decline of Pax Americans may push the EU toward greater coordination and interdependence. The Union's own structure ensures that cooperation will always remain a matter of compromise rather than automatic unity but is subject to change when the interests of European countries are forced to align.

THE IMPLICATION OF US RECIPROCAL TARIFFS

The return of reciprocal tariffs on February 20, 2026 from the United States changes both the economic infrastructure and the political logic behind global cooperation. In the short term, tariffs raise costs for exporters, disrupt supply chains, and make firms more cautious about investing. In the long term, they weaken trust in the American-led world order built upon stability and give the EU more opportunity and reason to build self-reliance through deeper internal cooperation, stronger industrial policy, and more resilient supply networks.

For decades, the American-led order rested on the idea that the U.S. would support open markets, predictable trade rules, and broad economic stability. Reciprocal tariffs challenge that system because they show that trade deficits can be used as pressure to counteract trade surpluses of other countries and boost domestic industry competitiveness. This protectionist measure serves as a warning sign for countries all around the world, encouraging countries to begin to diversify their investment in America. In the case of the EU, member states participate more in regional cooperation, domestic production, and alternative trade links. In other words, tariffs do not just raise prices, they also change expectations of country partnerships. Once businesses and governments expect future tariff shocks, they stop planning on the assumption that global trade will stay open and stable.

For the EU, the most obvious effect of tariffs is on exports. Reuters reported in April 2026 that EU exports to the United States fell by 26.4% in February, while the EU's trade surplus with the rest of the world shrank by 60% (Reuters, 17 Apr. 2026). That is a large decline, and it shows that tariffs are not a small policy adjustment but a real shock to European trade performance. The European Parliament also noted that higher U.S. tariffs and ongoing trade uncertainty have weighed on EU economic activity, with some countries and sectors facing much larger risks than others. This is due to the fact that some member states are more export driven and sector reliant than others. For example, smaller and more trade-dependent economies such as Ireland, Belgium, and Luxembourg are particularly exposed because they rely heavily on foreign investment, external trade, and the stability of the single market. As a result, they have less capacity to absorb tariff-related shocks independently and are often more dependent on coordinated EU responses. This uneven impact signifies how the tariff shock does not hit "Europe" in a single way. Some member states lose more sales, others face higher input costs, and some are affected mainly through supply-chain links rather than direct exports (European Parliament Research Service, 2026).

Tariffs also affect supply chains in a more hidden but equally serious way. Modern European production depends on long chains of suppliers across borders. A German manufacturer may rely on parts from the Netherlands, chemicals from Belgium, and logistics through ports in Spain or France. If tariffs make transatlantic trade more unstable, firms

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start to rethink where they source materials, where they sell products, and whether they should move production closer to final markets. This matters because tariff-driven uncertainty does not only reduce trade volumes; it also reshapes where Europe invests, produces, and competes, thereby affecting the EU's long-term economic cohesion and strategic autonomy. An article from think.ing also warned that the U.S. tariff shock is likely to reduce EU exports by more than 4% in both 2025 and 2026, and that the new tariff environment makes Europe less competitive in the U.S. market (Ruben Dewitte, 2026). Over time, that can push firms to reroute supply chains, shift investment, or avoid making new commitments in Europe if they fear weaker access to the U.S. market.

Investment is another major channel of impact. When tariffs are unpredictable, companies delay long-term decisions. They may hold off on building new factories, expanding logistics networks, or entering new partnerships because they cannot tell what the trade environment will look like next year. The European Parliament report says trade-policy uncertainty and tariff developments are expected to dampen euro area growth by around 0.5 percentage points in 2026–2027 (European Parliament Research Service, 2026). That figure matters because it shows tariffs are not just a temporary business problem. Rather, they can reduce growth, confidence, and productivity across the whole euro area. This creates a trickle-down effect: firms invest less due to anticipated declines in profits and rising costs, resulting in slower business expansion, fewer job opportunities, and weaker innovation. Over time, lower investment reduces the economy's productive capacity, making it more difficult for the euro area to sustain growth, remain internationally competitive, and recover from future economic shocks.

These short-term effects feed into a longer-term political shift. The more the U.S. uses tariffs, the more Europe sees the need to protect itself through strategic autonomy and self-reliance in the sense of producing strategic technology like missile defense systems themselves. The European Parliament's study on strategic autonomy says that economic security has become a top priority because of supply chain shocks caused by tariffs and that Europe must build resilience in impactful areas such as defense production for both competitiveness and security, energy supply, semiconductors, and critical raw materials (European Parliament Research Service, 2026). Currently, the EU is extremely dependent on the United States for these crucial areas because of their difference in industrial capabilities and their historical relationship with the country. This does not mean Europe wants to cut itself off from the U.S. The logic is simple: if Europe depends too much on outside powers, then every tariff, sanction, or political dispute becomes a threat to the supply chain security. A good example is the EU's Security Action for Europe defence instrument. The Commission has used SAFE to support joint defence procurement and to help member states buy equipment together instead of separately (European Commission, 26 Jan. 2026). This matters because cooperation reduces fragmentation between countries over whether or not joint defence between member states should occur. If countries each buy their own systems, Europe stays dependent on outside suppliers and misses opportunities to build scale. If they buy together, they strengthen the European industry and make their defence base more efficient. That is a concrete example of how external pressure, including tariff pressure and wider geopolitical uncertainty, can push EU countries to cooperate more closely. Another example is the EU's push to deepen the single market and improve capital markets. Reuters reported that the European Commission is attempting to remove long-standing barriers to cross-border investment and deepen EU capital markets because Europe continues to export a significant share of its savings abroad, particularly to the United States. Despite possessing high levels of household and institutional savings, European capital markets remain fragmented along national lines, with different regulatory frameworks, taxation systems, insolvency rules, and financial practices discouraging investment across member states. As a result, investors often find it easier and more profitable to place capital in the larger and more integrated

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U.S. financial market. This has led to a situation in which European savings help finance American innovation, technology firms, and economic growth rather than supporting investment within the EU itself.

The destabilisation of the transatlantic environment through reciprocal tariffs and growing uncertainty over U.S. economic leadership has made this dependence increasingly problematic. If access to the American market becomes less predictable, Europe risks becoming vulnerable not only through trade but also through its reliance on external financial destinations for investment returns. Consequently, the Commission's efforts to strengthen the Capital Markets Union are not merely economic reforms; they represent an attempt to retain European capital within Europe, improve access to finance for European firms, and reduce strategic dependence on an increasingly volatile external partner. From this perspective, the EU is not simply reacting defensively to tariffs but is actively seeking to build stronger internal economic structures capable of supporting long-term resilience and strategic autonomy (Reuters, 17 Apr. 2026).

In the long term, tariffs may actually speed up a reorganization of the global economy so that modern day economies can hedge against geopolitical risk and US protectionism. Instead of one dominant American-led system, the world may move toward a more fragmented order made up of regional blocs and strategic partnerships. Ignacio García Bercero, a senior fellow at the institute of Brugel who specializes in trade policy, argues that Europe should respond by strengthening the single market, investing more across borders, and building coalitions with other economies that share its rules-based outlook, focused on reduced dependency on one singular outside power. Europe wants to remain connected to the world, but on terms that make it less vulnerable to U.S. political swings (Centre for Economic Policy Research).

The effect on collaboration within the EU is therefore very important. Tariffs create pressure that can bring member states closer together because they show how much they gain from acting as a group. At the same time, the response is not easy because member states do not all want the same thing. Export-heavy countries may want open markets and fast deals, while others may prefer stronger protection, more subsidies, or more national control. Some states want common debt and joint procurement; others worry about sovereignty and budget limits. This means the EU's cooperation is likely to become deeper. Countries may work together more in areas where joint action clearly improves security or competitiveness, but internal disagreement will still limit how far integration can go.

LIMITS OF COLLABORATION

PRACTICALITY: CAN EUROPE DETACH FROM THE AMERICAN-LED ECONOMY?

It is not realistic for the EU to fully detach from the American-led economy in the short term. The U.S. remains one of the EU's largest export markets, and the Eurostat and European Parliament data show that the American market is central to Europe's trade balance. In 2024, the U.S. accounted for about 21% of EU exports by destination, which makes it the EU's single largest export market outside the block itself (ING, Ruben Dewitte, 2026). That means any sharp break with the U.S. would immediately hurt major European industries, especially cars, pharmaceuticals, machinery, chemicals, and industrial goods. A better way to describe Europe's position is not "detachment," but "partial de-risking." Europe can reduce its dependence in some areas, but it cannot quickly replace the U.S. market with another market of the same size and wealth.

That is why the tariff shock matters. According to Ruben, EU exports to the U.S. fell by more than 4% in both 2025 and 2026, and U.S. tariffs reduced Europe's competitiveness in the American market. Reuters also reported a 26.4%

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drop in EU exports to the U.S. in February 2026, showing that the pressure is already visible in trade flows (ING, Ruben Dewitte, 2026). These figures show that Europe is not in a position to “walk away” from the U.S. economy without real damage. Instead, the EU is being pushed to find ways to self-sustain itself while still staying linked to the U.S. market.

UNITY: DISAGREEMENT BETWEEN MEMBER STATES

A major obstacle to economic collaboration is that the member states do not experience U.S. tariffs in the same way as a result of the different priorities of governments. Germany possesses disproportionate influence within the EU because of its large economy, strong industrial base, and central role in European export networks. Consequently, it tends to prioritise market access, supply-chain stability, and the competitiveness of its manufacturers. France, by contrast, exercises influence through its diplomatic weight, military capabilities, and long-standing support for strategic autonomy, leading it to favour stronger industrial policy and state intervention. Poland has also become increasingly influential because of its population size, regional position, and growing security importance, but it continues to rely heavily on EU coordination and financing. These differences mean that the same tariff shock creates different political reactions inside the EU. As a result, member states may struggle to agree on a common response, reducing the EU's ability to act cohesively in trade negotiations and economic policymaking. This weakens the Union's collective bargaining power, creates policy uncertainty, and can limit the effectiveness of measures designed to protect European industries. Consequently, U.S. tariffs not only impose direct economic costs but also risk undermining economic coordination and political unity within the EU.

This divergence affects the process of cooperation. Countries that are highly export-dependent want quick and predictable trade deals. Countries that prefer more state intervention want the EU to use tariffs, subsidies, or industrial policy to defend its own markets. Countries that fear losing sovereignty are less willing to support deeper centralization. As a result, even when all member states agree that the U.S. tariff turn is a problem, they do not agree on the same solution. That is why unity in the EU is usually selective. The Union can often agree on broad goals, such as resilience or strategic autonomy, but it struggles when it has to choose specific goals.

TIMEFRAME: WHY AGREEMENTS ARE LONG TERM

This slow pace matters in a tariff crisis. Tariffs move fast, but EU law moves slowly. If the U.S. changes trade rules quickly, Europe has to respond through a system built for compromise rather than speed. That can make the EU look weak, but it also protects member states from being overruled too easily. The problem is that slow decision-making can be costly when businesses need clarity now. Firms need to know whether tariffs will last, whether subsidies will change, and whether new trade routes are being built. So while EU cooperation is possible, the timeframe often makes it less effective than a single national response would be.

WHAT COLLABORATION HAS ALREADY HAPPENED

Recent events show that collaboration within the EU is no longer just an idea; it is becoming concrete. One of the clearest examples is joint defence procurement, where the EU has expanded instruments such as SAFE, EDIRPA, and EDIP to encourage member states to buy military equipment together rather than separately as a form of increasing collaboration towards collective safety. This matters because the European Defence Agency has long found that collaborative defence spending remained far below EU targets, with only 18% of defence investment taking place

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collaboratively in 2022, even though the benchmark had been 35% for years. The first wave of SAFE funding, approved for eight member states, shows that these tools are now producing actual financial commitments. (European Commission, 26 Jan. 2026)

Another major case is the France-Germany joint declaration in 2026, which created a high-level nuclear steering group and committed both states to deeper cooperation on deterrence, early warning, air defence, and deep precision strike. The agreement is particularly significant because Germany's economic leadership and France's military and diplomatic influence often shape the direction of wider European integration. Therefore, their cooperation can influence the direction of European security policy more broadly. The EU has also used joint procurement in other crises, including COVID-19 vaccines, gas supplies, and ammunition for Ukraine, showing that crisis pressure can turn shared vulnerability into shared action. (European Parliamentary Research Service, 2026)

In energy policy, however, the pattern is more uneven: some states have embraced diversification and energy-community targets more fully than others, which shows that EU cooperation often advances through a common framework but uneven national implementation (European Parliamentary Research Service, 2026). Together, these examples show that the EU is already cooperating more in response to external pressure, but the cooperation is selective, uneven, and concentrated in sectors where states see direct strategic value.

LONG-TERM EFFECTS ON COOPERATION

In the long run, reciprocal tariffs may weaken the American-led world order by making the U.S. look less reliable as the anchor of open trade. If partners believe the U.S. may use tariffs again, they will plan for a future with more uncertainty and more fragmentation. That does not mean the U.S. loses all power, but it does mean that its leadership becomes less trusted. For Europe, that creates an incentive to rely more on itself and less on external stability. Over time, this could lead to a more coordinated EU economy, more defence integration, and more efforts to keep production, finance, and investment inside Europe.

But the long-term outcome is not guaranteed. The EU's deeper collaboration depends on whether member states can accept more shared responsibility. Germany may want competitive openness, France may want strategic protection, Ireland may want market access, and Poland may want security guarantees and flexibility. If those priorities clash too much, the EU may only achieve partial cooperation. If they align, the Union could build stronger resilience and more economic independence. That is why the real outcome of the tariff shock is not full detachment from America, but a slow shift toward a more self-reliant and more internally coordinated Europe (Reuters, 17 Apr. 2026).

CONCLUSION

The EU cannot fully detach from the American-led economy in the short term, because the U.S. is still a major export market and a central part of Europe's trade system. At the same time, U.S. reciprocal tariffs are pushing Europe to rethink dependence, strengthen internal collaboration, and build self-reliance in strategic sectors. The main limits are economic differences between member states, political disagreement, and the EU's slow decision-making system. But the main result is clear: tariffs are making Europe more aware that cooperation inside the Union is not just a political ideal, but a practical need.

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FROM POLITICS TO ECONOMICS: EVALUATING THE ORIGIN OF MODERN JAPANESE NATIONALISM

WRITTEN BY LEYAN JI



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ABSTRACT

The resurgent domestic popularity of the Japanese right-wing nationalist ideology in its local political landscape since the late 1990s can be seen as a manifestation of both economic dissatisfaction that existed within the population, and the pressure that originated from the escalating domestic frustration and tension between Japan and its East Asian neighbours including China and North Korea. The latter could be deemed more immediate due to the responsive nature and the potential national security risks for the country. Yet the primary manifestation, which is the united national identity and the desire for a strong nation, remains ultimately less tangible to the domestic electorate. This is due to the fact that geopolitical concerns are fundamentally political rhetoric in the media, while economic stagnation directly and disproportionately impacts the citizens. Concurrently, there are other supplementary influences, such as the revisionist approach to education, as well as the negative redirection of xenophobic rhetoric and populist narratives in media towards neighbouring states after their rapid development in the 21st century. Yet ultimately, the pivotal element of this debate is still the realist concern of economic burdens, which fulfils the crucial threshold of relevance and tangibility for the majority of the Japanese population. Which is due to its high degree of tangibility and direct impact on the living standards of the Japanese population, which is the closest, and arguably the most important determinant of public opinion.

INTRODUCTION

In recent decades, the global political landscape has witnessed a pronounced shift toward the revisionist, far-right alternative as compared to the more common pacifist, liberal political approach. As a country with deep-rooted connections to the establishment of nationalism and ultra-nationalism in the form of militarism, Japan acts as one of the leading figures in this process from the late 19th century up till date. This trend is generally interpreted as a form of modern reconstruction of the far-right ideology that has existed in Japan since the Meiji era (1868-1912). Although the trend itself and the impact are undeniably profound, existing literature suffers from a critical deficit: it rarely offers an explicit comparison between the factors that are widely recognised as main contributors to the shift. This paper aims to address this by tracing and evaluating existing arguments in academia, focusing on extracting the key comparison between the two largest facilitators to the change seen in the current society. These examine the economic factors seen in the widespread dissatisfaction of its people and the rising pressure regarding both tension in foreign relations and domestic political turmoil. The importance of cause in the field of politics is significant in a unique analytical lens, as the evaluation of these often becomes the foundation of theories such as the International Relations theories, which interprets societal systems and provide insights on patterns that helps us to understand the field of deontology further.

The resurgent domestic popularity of the Japanese right-wing nationalist ideology in its local political landscape since the late 1990s can be seen as a manifestation of both economic dissatisfaction that existed within the population, and the pressure that originated from the escalating domestic frustration and tension between Japan and its East Asian neighbours including China and North Korea. The latter could be deemed more immediate due to the responsive nature and the potential national security risks for the country. Yet the primary manifestation, which is the united national identity and the desire for a strong nation, remains ultimately less tangible to the domestic electorate. This is due to the fact that geopolitical concerns are fundamentally political rhetoric in the media, while economic stagnation directly and disproportionately impacts the citizens. Concurrently, there are other supplementary influences, such as the revisionist approach to education, as well as the negative redirection of xenophobic rhetoric

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and populist narratives in media towards neighbouring states after their rapid development in the 21st century. Yet ultimately, the pivotal element of this debate is still the realist concern of economic burdens, which fulfills the crucial threshold of relevance and tangibility for the majority of the Japanese population. Which is due to its high degree of tangibility and direct impact on the living standards of the Japanese population, which is the closest, and arguably the most important determinant of public opinion.

Many existing studies have extensively analysed the causal mechanisms behind this change, which is widely identified as a mixture of political pressure, domestic discontent regarding diplomatic policies and economic strategies. Current literature categorizes these elements often at the same level, and are often discussed separately. While some papers prioritise on diplomatic issues, others often highlight the structural economic stagnation, high deflationary pressure, and declining real wages in Japan. This created an active academic debate on the primary driver that is unsettled, while having it being fundamental in understanding states that are capable of such political shifts.

For instance, sovereignty disputes in the South China Sea between China and Japan, such as the Senkaku Island incident, are highlighted to prove the growing presence (3, 10), and projected as evidence of Chinese aggression by scholars. This is then utilised to justify the necessity of Japan having to revise its pacifist security posture, notably Article Nine, due to their geographical approximate positions and historical legacies. The disputes are utilised as external threats, which urges the Japanese population to unify under the same umbrella of nationalism that is capable of protecting the state. This then branches into two aspects regarding the role of the USA. Some argue that Japan must further increase the extent of its collaboration with the USA as a robust, primary source of military deterrence to potential threats of Japan's sovereignty (1, 5), while others suggested that as the USA is withdrawing from the global hegemon role and shifting into an isolationist actor, Japan is much more willing to develop its own autonomous military capabilities to counteract potential regional aggression from its Asian neighbours such as from China and North Korea (1, 12, 13). This friction is further illustrated in that although Japan was a large global donor of capital, its international recognition is poor, resulting in significant domestic backlashes against the checkbook diplomacy to gain recognition, and ultimately leading to Japan decreasing its presence in international institutions such as the Korean-led Northeast Asia Peace and Cooperation Initiative (NAPCI) that aims to establish order (11, 12). Nationalists often weaponise the public disapproval of the spending pattern, approaching it as an opportunity cost for domestic development, further fueling the discontent on the state's economic policies.

Domestic factors are another recurring concept in existing academia. For example, a stance amongst Japanese citizens argue that since the end of World War Two, the perceived, overly apologetic attitude of the postwar Japanese government undermined the pride of Japan as a country with its unique and significant cultural significance. This issue is exacerbated by the demographic of the Japanese population, as the generation that experienced the terror of war has mostly passed, and the younger generation often lacks the fear and reverence of war but instead glorified the martial traditions. This, and other factors such as xenophobia against Chinese and Koreans provided the base for nationalistic sentiments to manifest at a large scale amongst the Japanese populace. This is further complemented by the lowering of voting age in the Abe administration, a far-right nationalist politician that served as Japan's Prime Minister famously from 2012 to 2020 (3, 4, 11). This is accompanied by the structural revisionist change in action, such as the revision of education materials and increasing frequency of leaders' official visits to the Yasukuni Shrine, in which Class A war criminals from World War Two are honoured. All of which are diplomatically disadvantageous,

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but morally significant as a fuel to the nationalistic sentiments in the Japanese population (4, 11).

Last, the economic failure of Japan in the past few decades was also frequently discussed, and can be summarized to a few elements. For instance, some scholars noticed that the economic structure of Japan itself is highly fragile and susceptible to external impacts, namely the significant portion of workers in small firms which are immensely protected, subsidized, yet is eminently unproductive and uncompetitive globally (3). In addition, the culture of seniority being more weighty than merit also suggested that the economy is unlikely to be flexible in response to the current economic environment (8). This is then expanded by many scholars to explain the economic failure of Japan, with the most evident example being the “Lost Decade” and the disastrous impact of the 2008 world economic crisis, all of which leading to the significant downfall of Japan’s economy, resulting in a large gap between now and the golden era of Japan when it was the second largest economy in the world. It is argued that both the reducing living standards and the relative deprivation of economic power contribute to the increasing nationalistic sentiments, which politicians utilised to attract voters with reforms such as the “Abenomics”, as well as the psychological returns for the lost financial wages through embracing nationalistic cultures (1, 3, 4, 8, 15).

While all of the existing studies provide valuable context to the issue, gaps still exist in terms of the evaluation of these causes. In most of the research, all the elements are given equal weight as to their contribution to large-scale nationalism. Although it is true that they are all valid contributions, the fragmented evaluation and analysis of separate parts can be potentially misleading as to how the wave grew to such an extent and scale. By reorganising the two largest components of all the causes, it becomes possible to discern how the Japanese population perceived these issues differently. The nuance in the individual agency aims to offer a more holistic analysis of the demographics, which could potentially provide the foundation for the evaluation of wider political case studies that relates to the development of nationalism.

THE NATIONALIST IDEAL IN JAPANESE POLITICS

In the late 20th and early 21st centuries, political turmoil seemed to be the defining characteristic of Japanese politics in both demographical and diplomatic aspects, leading to large public discourse. This eventually developed into sentiments of nationalism, especially amongst the youth of the time. While frequently cited in academic literature, the tangible impact of these political shifts on the average citizen remains a subject of debate. This section evaluates both domestic and external factors in Japan’s political landscape with focus on three key criterias: scale, profundity and political response. By applying these criteria, we will be able to see the significant role of politics in daily life. While these political rhetoric may sound compelling and high on scale, the relative tangibility and degree of impact is largely subservient to economic factors.

THE REVISION OF HISTORICAL REMNANTS

The potential for Japanese large-scale nationalistic development is undeniable in Japan, as nationalism has persisted since the late Tokugawa era (1600-1868), and later developing into a nascent version of the modern one in the Meiji era that followed. The original form of nationalism aimed to develop Japan as a whole, after the emperor saw the large disparity between Japan and Western nations due to their industrial revolutions and development of technologies. After generations of effort to develop, the outcome was that Japan at the time prospered and became one of the strongest nations in Asia. This very much fed into the ego and pride of the Japanese populace, and fueled

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the idea that Japan as a country held superiority over other lesser developed nations. One example of this would be the concept of the “Great East Asia Co-Prosperity Sphere” during the Second World War, in which Japan sees itself as the leading figure to bring up its undeveloped neighbours. One other crucial aspect of the original ideology is that Japan is a sacred country that is protected by the sun goddess Amaterasu with the emperor being the descendant of the goddess herself, which further enhances the idea that Japanese people are uniquely superior compared to all others. (Wikipedia, Japanese Nationalism) The implication here is that the ideology of nationalism and Japanese superiority is deeply integrated into the education of new generations more or less as part of the country’s identity. This is seen in the school enforced cultural activities and the curriculum that often had an emphasis on past economic or military glories. This inherited ideology, alongside the “Japanese miracle” period from the 1960s to the 1980s together crafted the picture of a strong and powerful nation that is independent. The key inference here is that the divine identity made nationalism in Japan especially convincing, as a fundamental shared belief that is part of the nation’s culture itself. (Matthews, 2003)

Still, the Japanese population has demonstrated a susceptibility to the populist narrative of reclaiming lost national prestige after militarism in Japan was eradicated by the world order after World War Two. Parallels can be drawn from the past with the present, as one of the key actions of the Abe administration was to revise the school curriculum, utilising it as a sanitized national rhetoric. One example would be references to “Comfort Women”, which refers to the captured women and girls that are forced into sexual slavery by the imperial army during the Second World War. This revisionist approach largely fostered the growth of emotions such as Japanese pride and superiority, all of which having a large potential of developing into nationalism. (Koichi, 2016) Patriotic beliefs, altered history, and xenophobic attitudes integrated as parts of education can be extremely penetrating and persuasive to the younger generation, who typically lacks exposure to a diverse range of perspectives from the international field due to Japan’s generally isolationist approach in cultural matters and media. A voter base at the scale of a generation with patriotic and xenophobic deeply implanted in their minds offers the most potential for nationalism to grow in the country.

THE INTERNATIONAL STAGE, APOLOGETIC DIPLOMACY, AND CHANGING PERCEPTIONS

During the late 20th and up until the 21st century, Japan was a leading donor on the world stage, and this even included an ODA (Official Development Assistance) investment in China, in which Japanese people originally took pride as a means to shape a prosperous Asia. This was a very substantial support, as the investment allowed the secondary sector in China to rapidly develop, which then boosted the Asian economy as a whole due to the increase in trade. (Matthews, 2003) In addition, their contributions in international institutions such as the UN and the World Bank are extensive as well, this includes being the second-largest capital supporter of The World Bank. However, the lack of Japanese figures in high positions of these institutions, and the lack of recognition from China and Korea regarding their apologies for wartime crimes suggested hinted at the fact that their financial contribution yielded negligible diplomatic returns. This situation is more evident in the 1990 Gulf War against Iraq, in which Japan had sent US\$13 billion worth of support, but received no recognition from Kuwait (Wallace, 2016) other than criticisms regarding Article Nine in the Japanese constitution, which renounces Japan’s right of war and maintaining military forces. The disparity between contribution and influence was a major factor in Japanese people not recognising the ideals of pacifism which the rest of the world celebrates.

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The idea here is that a large portion of Japanese people are unsatisfied with the return they receive from their financial investments for diplomacy. At this stage, the Japanese population's perception of soft-power diplomacy became that it failed in achieving the aim of regaining international recognition. It is then natural that some pivoted to the harder stance of nationalism that promoted military strength. (Wallace, 2016)

This coincided with the “apologetic” diplomatic strategy the Japanese government used towards China and Korea for its war crimes, seen in leaders avoiding official visits to the Yasukuni Shrine to abstain from diplomatic friction. The strategy is widely opposed by the young generation in the 21st century, as they perceived it as actively eroding the national identity, which, as mentioned earlier, was a crucial part of their education. (Matthews, 2003) This overall resentment towards the unbalanced international recognition of Japan acted as the primary catalyst for the nationalist sentiments in Japan, as it didn't represent any specific individuals or groups, but it represented the Japanese identity they found pride in. This overarching suppression of Japan suggests that the scale of political discontent not only exists, but exists at a large scale in Japan. This is significant because as mentioned above, the Japanese identity is universal across the demographic apart from policies, and the subconscious distinction between the two made it extremely convincing even to the part of the demographic that is uninterested in politics.

In reviewing the policy of Koizumi and Abe, a key similarity is that they aimed to restore Japan's national prestige and sovereign stature instead of focusing on more desperate issues such as inflation or fewer innovations. This distinction reveals the key to their successful campaigns, which is that in such times of widespread dissatisfaction, a unified identity addresses a widespread desire for national cohesion. The prerequisite for this campaign strategy to work is the overwhelming scale of people who share similar thoughts, thus proving the argument that scale of political resentment was a key in the uprise of visible nationalism.

SOVEREIGNTY, THREATS, AND AGGRESSORS

Beyond the scale of the impact, profundity of political threats are equally appealing, mainly seen in the increasing tension between Japan and external actors. After World War Two, Japan, having been a loyal ally to the USA, acted as a stabilizing force in Asia as USA's outreach station. Japan has long relied on the bilateral defence relationship with the USA, but when North Korea started to conduct their nuclear tests in 2006, the domestic attitude towards the reliance started to change. The Japanese public grew increasingly intolerant of the persistent existential threat, and the best way for Japan to avoid the risk at the time was nothing other than the direct intervention of the US to eradicate any nuclear device in North Korea. (Atanassova-Cornelis, 2014) However, the US decided to not engage directly and instead adopted more isolationist policies towards them. The steep contrast between the potential harm to Japan and the inaction of the world resulted in a widespread strategic anxiety amongst Japanese citizens and the government, further pushing them towards the goal of remilitarising through the revision of Article Nine to protect Japan. This coincided with the divergence of US interest from Japanese interests, leading to more questioning in the security guarantee. For instance, In a 2003 poll by Asahi Shimbun, 60% of Japanese citizens thought the US led Iraq war was not justified. In addition, some also started to question the impregnability of the US after the event of September 11th, which further increased the necessity of an independent force of Japan. (Matthews, 2003) To make clear, Article 9 became part of the Japanese constitution following their defeat in the second world war, and states that Japan must renounce war as a sovereign right, and may not have a standing army with the potential to maintain war potential. Although the intention of North Korea remains unclear, their ability to destroy Japan is severe enough for its people to realise that the country must be able to defend itself instead of relying on external actors for security.

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(Woodward, 2017) Although the collaboration between Japan and the USA remained tighter than ever still, the public fear of Nuclear threat provided a sufficient rationale for the Japanese population to vote for the nationalistic ideals that promised protection. This is exceptionally powerful as political rhetoric, as the perceived existential threat is much stronger compared to other forms of diplomatic issues in the minds of average citizens. When the threat itself is intolerable, it is able to bypass certain political logic and catalyse the growing of more extreme forms of nationalism that is perceived as capable of securing the state. (Berkofsky, 2017)

Apart from the threat of North Korea, tangible events also played a crucial part in strengthening Japan's pivot towards rearmament and support for nationalism. The contested administrative history of the Senkaku Island dispute between Japan and China is the most well-known due to the large media coverage and lasting effects internationally. The island is located between the Okinawa island of Japan and Taiwan. The Island was supposed to be given back to China after Japan's defeat in the war, but the US, who held the island, gave Japan the administrative right of the island. In 2010, Japanese coastal guard captured a Chinese trawler in the water, and threatened to prosecute it under the charge of incursion into the Japanese Exclusive Economic Zone (EEZ). The ruling party, the Democratic Party of Japan (DPJ) released the trawler, and soon announced a large joint military exercise with the US in Japanese waters to deter China from further actions. The DPJ's softer approach towards China created a domestic backlash, which established the contrast between it and the LDP's stronger image and nationalistic narratives. (Woodward, 2017)

A significant escalation occurred in 2012, when Ishihara, the nationalist governor of Tokyo urged the Japanese government to buy the land deed of the island for resource extraction without notifying China. Subsequently, Chinese surveillance and warships are constantly seen in the surrounding areas, and the widespread anti-Japan protest rose across 100 Chinese cities, coinciding with the boycott of the International Monetary Fund (IMF) and World Bank meetings in the same year. As a result, tourism and export plummeted for Japan, yet Abe took a hardline pro-Japan stance, and used this to further justify the military presence of the US in Japan, increasing the military budget, and further revision of Article 9. Similar events happened to the Dokdo Island of South Korea, which also led to a series of boycotts and overall negative sentiments against Japan. The Japanese response here is not to de-escalate the events to avoid consequences, but to utilise the negative perceptions of China and North Korea to achieve their own goal of revising Article 9. This showed that the strategic aim for the responses was not to avoid any practical loss, but to strengthen the xenophobic perspective amongst the Japanese population. This is then utilised to achieve nationalist aims in the end, suggesting that it is nationalism that remains at the core of government's diplomatic responses instead of truly beneficiary incentives. (Woodward, 2017)

Those developments demonstrated that Japanese people's concerns about foreign aggression are empirically grounded. The reputation of Japan amongst its neighbours has traditionally been negative for historical causes and practical reasons like the incidents, while Japanese people find pride in them. As a result, the frequent "offences" by China is seen as active aggression against the rights and sovereignty of Japan as a country. This largely contrasted with the pacifist diplomacy China adopted in the past, and this also made Japan realise that China is starting to achieve strategic parity with them militarily. (Atanassova-Cornelis, 2014) The overall negative impression received by the Japanese population together with the extensive government propaganda provided the necessary drive for citizens to take a firmer stance against Japan's Asian neighbours in matters such as foreign policies. The negative sentiment then evolved into xenophobia in some instances, and support for nationalistic beliefs in others. In the 21st

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century, sales of patriotic movies and magazines skyrocketed in Japan. For instance, a film telling the story of Kamikaze pilots has a place in the top ten most successful films of all time (Wallace, 2016).

This opposition also coincided with the rapid development of China in the early 21st century in nearly all aspects, notably military technology like missiles, innovations in the internet field, and its economy, to the extent that Japan is being overtaken. The contrast between adversaries growing and Japan itself staying stagnant was considered even more serious in the younger generation that believed in the Japanese glory. (Atanassova-Cornelis, 2014) The buildup of fear and anxiety rose as an almost inevitable consequence of a multitude of reasons, including the perceived retrenchment of the US security guarantee, China rising in military and showing a clear intent of aggression, compounded by Japan's diminishing influence on the world stage. Nationalism at the time seemed like the only side that valued Japanese glory, especially when the alternatives either heavily promoted external support and international contribution as the solution to Japan's problems. The promises of a restrengthened military acted as the strategic framework for Japan to regain its right as one of the largest economies in the world when surrounded by threats that posed an existential risk to the state.

SOVEREIGNTY, THREATS, AND AGGRESSORS

Lastly, the development of nationalism is oftentimes reflected in the political moves made by the government. The chaotic political landscape has given rise to a few major policy changes that are notable, the first and most significant of them being the revision of Article 9. The call for revision always existed, but only peaked as a part of the nationalist agenda. The impact of the change is not only tangible, but more importantly representative. In 2000, 41% of the Japanese public were in favour of it, but the figure rose by 6% in only twelve months, highlighting the urgency of the matter within the minds of the Japanese populace. (Wallace, 2016)

The revision, to many other countries, while suggesting a regression toward Japan's pre-war militaristic posture, was seen as glorious and necessary in the perspective of Japanese citizens as a step towards its historical pride. Furthermore, this is bolstered by the prevalence of regional geopolitical aggression in Asia, making the momentum for change in foreign policy even stronger within the state. The significance of the change is that it signaled to the world that Japan no longer wanted to abide by the world order established post-World War Two, and wanted to be freed from the reputation of a defeated power, reclaim its normal statehood and sovereign rights.

While this argument is compelling, it was an extremely bold move for Japan, as this risked large scale sanctions from its neighbours that are historically rivals of Japan. Its consequence transcends beyond diplomatic posturing and diminishing international relationships with geopolitical neighbours, but is actively depleting the practical benefit Japan can get. (Woodward, 2017) Thus, potential consequences of not collaborating with China can be enormous, specifically considering its reliance on natural resources. At this stage, Japan's desire for an army was strong enough that they are willing to give up certain capital benefits, which is sufficient to prove their intent to become a conventional state. At this moment, nationalism is perceived to hold the highest likelihood of achieving this aim, and the electoral outcomes have proven their determination for positive change.

Examining Japan's strategic partnership with the USA reveals the uniqueness of nationalism as the solution. As mentioned before, Japan had been a traditional ally of the US, and this continued, if not since strengthened, under the administrations of Koizumi and Abe, although the inadequacy of support from the US already raised public

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discontent. (Woodward, 2017) From the divergent diplomatic postures, it is evident that the role of Japanese nationalism here was to foster othering within the Japanese people to ensure domestic unity rather than being a genuine pursuit of self-reliance. An example of this would be Abe's idea of "Proactive Pacifism", which ultimately still links back to connections with the USA. (Atanassova-Cornelis, 2014) This claim can then be used to prove the xenophobia that is targeted at Chinese and Koreans, as well as the neglect of potential retribution from China. As long as Japan stays under the protection of the USA, all those harms can be mitigated at least. From the bold intent to revise Article 9 itself, it can be argued that the aim of the new Japanese nationalism is to unify its people towards the common enemy in Asia and encourage deeper collaboration with the USA. Furthermore, it can be inferred that the move is largely utilitarian with the extensive political rhetoric targeting the narrative of existential threats, making it more compelling in the sense that it is the necessity for the state instead of a choice.

THE RECONSTRUCTION OF NATIONAL IDENTITY: WAR CRIMINAL OR NATIONAL HERO?

The shift in diplomacy is also seen in other actions of the government, such as the revision of educational curriculum and paying official visits to Yasukuni Shrine. Both are representative of Japan's diplomatic strategy shifting from pacifism to a clear anti-Chinese stance. The revision of education, as mentioned at the start, is largely denialistic of historical events that tarnish Japan's international reputation, which necessarily includes the reevaluation of contents such as massacres and human experiments during the second Sino-Japanese war. This was extremely problematic in terms of the potential controversies and tension it brings to the nation. While pedagogical reforms target the next generation, official visits to the Yasukuni Shrine serve as a more immediate, symbolic assertion of sovereignty. The modern purpose of the shrine was to commemorate Japanese officials who died in wars, including those who are accused of being Class-A war criminals in World War Two. The place is perceived as a legitimate site of veneration and is sacred in the eyes of hardline nationalists, such as Abe himself. Yet, paying official visits frequently leads to discontent from China regarding the sensitivity of the shrine. (Koichi, 2016) The explicitly shown political acts ultimately bypass the conventional limitations of law and deteriorating international perception, and redefines the cultural and spiritual significance of its history. Through honouring war criminals, the state essentially challenges the postwar pacifist order that labelled them as criminals. This radical reconstruction of societal perceptions was not only significant, but also politically unprecedented.

However, the shrine was fundamental in demonstrating the state's resolve to safeguard Japan's heritage in the eyes of the population. For instance, a 2014 Asahi Shimbun poll suggested that 60% of people in their thirties supported government officials' public visits to the shrine. (Wallace, 2016) Both Koizumi and Abe leveraged their visits as a strategic opportunity for electoral mobilization, which was a significant departure from the diplomatic caution exercised by their predecessors. Ultimately, the controversies surrounding education and Yasukuni represented a political leader's dedication for an independent Japanese identity that is not afraid of external oppression, and utilising the thereby cultivated, favorable public image and their strong association to nationalism to shift societal opinions. The subject of all three examples mentioned are highly sensitive yet representative of the Japanese identity, hinting at the subtle transformation of the concept of "Japanese identity" itself, transforming it into what nationalistic politicians intended. Overall, Japan has shown great political incentive through revisionist actions to depart from China and other neighbouring countries as a response to their "aggression". It can be inferred that the unprecedented political response was not only a radical departure from the conventional framework of diplomacy and politics, but an active reconstruction of the Japanese identity. The strategic implication of this is that nationalism will be integrated as part of the social framework in Japan in the long term, instead of a solution to only solve

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immediate issues Japan faces.

Throughout the contemporary political landscape of Japan, political discourse and state initiatives were largely centered around the concept of “Japanese identity”, to suggest that Japanese people must form a unified national consensus to achieve the end goal of a strong and independent nation. With an overwhelmingly strong sense of fear for a perceived existential and overwhelming regional threat, it is not surprising that Japanese citizens are willing to believe in the side that is firmly pro-Japan and demonstrated their loyalty with radical steps towards the goal.

THE DECLINING JAPANESE ECONOMY: SIGNIFICANCE OF MATERIAL FOUNDATION OF PUBLIC DISCONTENT

The importance of economic development and stability is a primary indicator of a strong country, while this has been stagnant in Japan from the late 1990s till date. Japan has experienced a multitude of major economic contractions in the past few decades right after the Japanese economic miracle, leaving a substantial emotional gap within its population. The two most visible contributors were the 2008 global economic crisis and the “Lost Decades” of Japan’s economy beginning from the 1990s until the 2010s. In both situations economic stagnation became most acute for the Japanese economy and its population. While those events were substantial in accelerating the recession, the structure that the Japanese economy operates on is already fragile and inefficient, susceptible to external shocks for various reasons such as aging and valuing seniority over merit due to the complex nature of the economic elements. (Aoki, 2013) This section explores their significance with the metric of scale, profundity and political responses, aiming to clearly differentiate the core of how separate causes ultimately catalyze the growing nationalist sentiments amongst the Japanese population. As a summary, the tangible pressure in the financial situations of Japanese households, as well as the rigid economic structure was primary in evoking and transforming the public discontent towards the support of nationalist solutions, with the impacts on them acute and tangible.

STRUCTURAL ISSUES IN THE JAPANESE ECONOMY

One of the most fundamental analyses here regards the scale of economic impact on the population. This is especially prevalent in the context of Japan, a country that historically predicated its national prestige on economic prosperity. Japan as a country had suffered persistent economic stagnation through the “lost decades” (late 1990s to the early 2010s), at the same time experiencing major domestic crises and the 2008 economic crisis. Undoubtedly, the length of economic depression for an advanced country is virtually unprecedented among contemporary advanced economies. Apart from the timescale, the demographic scope of this impact is equally significant for a few reasons that lie in the economic structure of Japan. For instance, the majority of Japanese firms are protected by the government in various ways. This includes substantial amounts of subsidies for small and medium-sized enterprises (SMEs), as well as policies such as the 1949 Foreign Exchange Law and the 1950 Foreign Investment Law that together imposed a restrictive and state-controlled structure for Foreign Direct Investment (FDI) into Japan, aimed to stabilise the economy. From this, inference can be drawn that recessions and external shocks will likely be felt by the entire economy instead of limited to certain sectors, establishing the fundament for the unconventional yet universal backlashes felt by most of the population. (Woodward, 2017)

With this said, these firms often lack the crucial competitiveness and flexibility relative to other countries due to the low competition pressure domestically which takes away their incentive to maintain a high degree of efficiency. This

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overall lack of positive change for most domestic corporations resulted in the vulnerability of them against external shocks such as the rising productivity of China overtaking, to say nothing of global macroeconomic shocks. Furthermore, the sector distribution of workers in Japan is also highly inefficient as a whole. Some examples include the 4% of all workers that work in agriculture as compared to the 1.4% of the US. This is complemented by the 72% manufacturing employees concentrated in small-to-medium enterprises (SMEs: firms that are less than 250 people) instead of large-scale, highly efficient corporations. These small firms relying on labor-intensive rather than capital-heavy methods in Japan are arguably much higher compared to the other developed economies such as the USA or Western nations like Germany. The inefficient allocation of human resources in Japan ultimately means that when the overall economic situation deteriorates, the primary burden falls upon the workers themselves. Furthermore, the system lacks efficient measures to reintegrate older workers into new industries, further widening the gap between Japan and other advanced countries, while leaving even more burden on the younger generation and the workforce overall. (Woodward, 2017)

The scale of dissatisfaction underscores the profound psychological impact onto the Japanese population, to the extent that they are much more willing to embrace large reforms and fiscal stimulus. Unsurprisingly, over 90% of Japan's exports consist of highly income-elastic goods, typically consumer durables such as cars and electronic appliances. At the same time, Japan as an island nation is also reliant on imported energy. It can be inferred that the stable income of the manufacturing sector in Japan is not only highly reliant on demand that is typically from Western countries, but also on the stable supply of energy. When the 2008 world economic crisis hit, both elements were taken away, and the Japanese economy experienced a severe contraction. In the first quarter of 2009, revenue from exports fell by 36.8% alone. This acutely worsened Japan's manufacturing industries, such that a prolonged period of stagnation across the entire manufacturing sector was precipitated after 2009. (Kawai and Takagi, 2007) The widespread economic precarity of the Japanese workforce during the time acutely allowed nationalist politicians to increase their appeal for a strong Japanese economy. When the 2008 Economic Crisis sufficiently proved the failure of post-war liberal order in stability, nationalists had a much stronger ground in arguing the necessity for Japan to be self-reliant. Which had directly led to the later popularity of policies such as Abenomics that proposed significant reforms.

EVALUATING THE WORKING DEMOGRAPHY

To start with, Japan's aging population problem is one of the most severe in the world, which indicates a contraction in the labor force and a sustained decline in aggregate economic output. The implication of this is that the dependency ratio is projected to simultaneously rise significantly, leaving more burden on future generations as a whole, while the low fertility rate remains constant. The consequence of low fertility rate is likely a decrease in output and lower wage rate due to increased cost, while less will be willing to give birth to children when their economic situation is stressful. This negative cycle of low economic growth and low fertility constitute a nationwide issue, and acts as a fundamental threat to the long-term vitality of the state.

On the ground of the working demography, systemic issues potentially exist at a greater scale. Due to the overwhelmingly high prevalence of intergenerational family businesses in Japan (Morishita, 2022) and its protectionist economic policies, traditional and conservative beliefs are institutionalised within such firms. Examples of this would be the ideal of "lifelong employment", and valuing seniority over merit when assessing workers. This structural rigidity is particularly detrimental given Japan's rapid demographic aging in the 21st century. Thus, older

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workers oftentimes have much less occupational mobility compared to the young generation. An inference here is that the conservative and traditional values these firms held made them the natural voting base for nationalists who put the Japanese identity at the core of their campaign. (Aoki, 2013)

The institutional response for this after the 2008 recession was a significant labor market deregulation and erosion of employment protections, leading to much less stability in the income of those that are young. This shift was characterized by the proliferation of short-term contracts for young workers while maintaining the lifetime security of senior employees. Yet, this only exacerbates the intergenerational inequality in the long term. Taking into consideration the remaining depression in manufacturing, this strategy resulted in three structural failures within the context of Japan's corporate landscape. (Aoki, 2013)

The erosion of human capital also is a prevalent issue amongst the Japanese workforce. Younger workers often endure difficulty in securing promotions, as higher positions are often kept to more senior workers. This eventually leads to difficulties in securing financial requirements and opportunities, especially when supporting a family. In addition, short-term contracts deinceivizes firms from investing in human capital, further reducing their ability to find desirable occupations afterward. Furthermore, less training of new workers leaves the firm no other option than to keep relying on senior workers with their skills that were built with experience. However, it is common for older workers to struggle to adapt their expertise to the production of high-growth modern commodities, and would instead stay with the original production that could easily be outdated. (Aoki, 2013) This mismatch between supply and demand for Japanese goods would almost certainly lead to a reduction in revenue, which exerts downward pressure on real wages. This problem is especially prevalent in the modern age due to the rise of China as the main export country with goods that are much more price competitive. The key here is that the economic stagnation is not a temporary issue, but a structural decline of the Japanese workforces' capability. This suggested that the impacts felt by the population are not only universal, but are also unlikely to be improved over time, leading to further degradation of output and economic growth. This nuance in the cause of national recession is exploited by nationalists by promising bold reforms and the return to past industrial glory.

The overall impact of the economic stagnation specifically for the younger generation can be summarized as an erosion of both real wages and the pathways to future earnings, and when both the skill level and wage decreases for younger generations, they're predisposed to seek political alternatives that promise to alleviate these systemic pressures with sufficient agency. The current workforce lives in the shadow of the Japanese economic miracle enjoyed by previous generations, and the sharp contrast in quality of life further reinforced the idea that Japan must do better, which nationalism is perceived to promote through its hardline stance on economic reforms and efforts in reducing foreign intervention.

THE OFFICIAL RESPONSE

The state's policy responses, most notably the Abenomics program initiated during Shinzo Abe's second term, provide a significant insight into the electorate's socioeconomic anxieties and aspirations of the Japanese electorate. Abe's economic policies consist of a series of economic actions (Three Arrows) that utilises monetary policy, fiscal policy and structural reforms. The monetary aspect focused on the program of Quantitative and Qualitative Monetary Easing, a group of actions that aims to achieve a 2% inflation rate over two years. This program involved the massive expansion of the monetary base, executed through the Bank of Japan by purchasing an extensive amount of

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Japanese Government Bonds to redistribute money, as well as increasing the purchase of exchange-traded funds (ETFs) and real estate investment trusts (REITs). The fiscal side was more straightforward through the direct increase of fiscal spending to stimulate economic activities as well as infrastructure revitalization. Consumption tax was raised from 5% to 8% in 2014, and is projected to increase again to compensate for government spending. Meanwhile, the government also wanted to reduce corporation tax to encourage hiring more workers. Lastly, structural reforms with various aims were implemented. This includes support for industrial regeneration, assisting the growth of SMEs and innovative technologies, the expansion of the labor force for the female, elderly, and the young, creating new markets in various sectors ranging from healthcare to fishing, as well as globalisation. While the full scope of structural reforms faced significant bureaucratic and political hurdles, they remained central to political discourse. (Akram, 2014) Although the monetary component of the policy is most frequently mentioned in studies, the rather bold fiscal and reform components were the real stimuli that did bring short term change to the Japanese economy in 2013 following its implementation. The proposal of Abenomics very much filled in the gaps of the nationalist narrative by providing the tangible strength Japanese electorates demanded. For instance, increased spending on welfare and infrastructure represented Japan's capability to rebuild national strength and cater to its people. Furthermore, structural reforms were framed as active revitalization of the nation as another way to prove the strength of Japan. By actively creating new markets and globalisation, Abe aimed to prove to the Japanese electorate that Japan has the potential to stand against China, which accurately matched the political narrative and the psychological needs of the Japanese populace.

The implementation of these policies yields two primary insights, the first being that the importance of the issue is undoubted for the Japanese voters for Abe to use it as the core of his campaign, highlighting the relevance of those issues to people's life. The second is the emphasis on productivity, which was much less catered to previously. While the overall merit of the policy remains controversial, the assertiveness and institutional resolve displayed in the process of installing bold economic policies is evident. Inference can be made that it represented a radical departure from the incrementalism of previous administrations to the Japanese population who suffered from deflation for two decades. (Akram, 2014) The widespread support for Abenomics illustrates a clear desire from the public for a functioning economy from all demographics, but especially amongst the younger generation, who bore the disproportionate weight of structural stagnation. By framing economic policies as a collective effort of Japan, Abe was able to incorporate the nationalist ideal for a strong state into the larger, universal demand for a functioning economy.

Through the analysis of different economic aspects, it is evident that the issue of economic recession from the perspective of the Japanese populace are not simply significant but existential, especially amongst the younger generation. Concurrently, the rise of nationalism as the solution to the overwhelming problem seemed to be the most valid and tangible alternative, seen in the rapid execution of Abenomics and the large public support. The frustration of Japanese people being systematically excluded from stable employment and wage growth were eventually expressed politically through the support of a leader that they believe will revive the glory of Japan. Amongst the economic desperation, nationalism provided the necessary ideological framework for the mobilization of the Japanese population.

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FROM PARLIAMENT TO PRICE: WHAT IS IT, AND WHY DOES IT MATTER?

When the two major factors are considered through a unified analytical framework, it may seem that they are equally valid and strong for the development of nationalist sentiments in Japan. Although this statement is true to some extent when taking into account the complex social structure and interplay of elements, this paper contends that the economic concerns Japan faces is the primary reason for the modern form of nationalism to grow to such an extent that it now defines the contemporary political landscape. The landslide victory of Sanae Takaichi, who claimed to be the predecessor of Abe as another far-right nationalist, in 2025 with nearing 80% support rate can sufficiently prove the nationalist's political dominance. With both means already proven to be key causes and accelerators to nationalist sentiments, some criteria can be used to evaluate the relative weight of these drivers. The main aspects to consider when assessing the significance should still be the tangible immediacy and systemic profundity of their effects.

ASSESSING RELATIVE SCALE OF ECONOMIC CRISIS AND POLITICAL DISORDER

On the matter of scale, where tangible impact is the primary metric for most, economic stagnation is much more relevant to the population of Japan for a few reasons. First, a defining characteristic of many advanced electorates is that a substantial portion of the population does not engage in politics. This implies that issues mentioned regarding sovereignty and external threats, may be limited to a specific subset of the population without leaving any foreseeable impact. Across the Japanese General Social Survey from 2008, 2010, and 2012, an average of 55.3% of the sample population either do not have a particular party that they support or know about. It is worth noting that these years are typically hard times for the Japanese population, with the large economic recession in between. These statistics illustrate the limited scale of active political engagement, which is the prerequisite for nationalist campaigns targeting foreign policies to be effective. (Osaka University of Commerce, 2008) (Iwai and Tanioka, 2010) (Osaka University of Commerce, 2012)

However, most people in society need to be in an occupation to live, and when both cases only consider adults, the economic variable demonstrates a significantly broader demographic reach. To make it even clearer, Japan is a country that relies heavily on both exports and imports, and the world economic crisis alongside domestic structural issues effectively curtailed the volume of Japanese exports within the relevant timeframe, which lasted until today. (Kawai and Takagi) This, alongside the large portion of workers that work in the manufacturing industry and are insufficiently protected from external shocks, provides a compelling empirical basis for the argument that the economic impacts were fundamentally impactful on the daily lives of the citizens. With this link made clear, it is more obvious that the scale of tangible impact on the economy is much more applicable and relevant compared to ones in the field of politics.

COMPARATIVE RELEVANCE OF FACTORS ON THE DEMOGRAPHY

External threats, although strong, are largely speculative and are based on predictions. Although this doesn't diminish the validity of the truth in it, the lack of quantifiable evidence is naturally less persuasive than more visible aspects of the same issue. There are two key perspectives to mention, the degree of impact and the demography that takes the impact. Although the potential harm of external aggression is prevalent, there are many drawbacks to the extent to which it can reach the minds of the local populace. For instance, although the incursion of China and North Korea in Japanese water represents the disregard for national sovereignty and significant danger, the impact of the event on

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the average citizen remains abstract unless filtered through state-led campaigns or nationalist propaganda.

The same applies to issues such as education and Yasukuni Shrine. Most important is that intense threat, or even direct danger does not inherently catalyse public mobilization; rather, without a direct link to personal welfare, such threats often result in civic inertia rather than political action. With this established, we can see that not only that the reach of political discourse is essentially restricted to a politically active minority, but also that the resonance of political shocks is mitigated as they filter down to the general populace.

On the contrary, economic turmoil is naturally relevant for the majority, if not all of the population, either directly or indirectly. The sharp reduction of export, rise of cost of production, and lack of innovation inevitably lead to the decline in economic growth, which consistently correlates with a decline in real wages. It is worth noting that economic decline can be cyclical, such that less wage leads to less spending, which then leads to reduction in aggregate demand, which leads back to less purchasing power of the workforce. Thereby, the process repeats itself. This is the key reason why governments in times of recession must encourage spending, evident in Abenomics itself.

In the debate over relevance, it is also worth looking into the demography itself, and identify the sectors of Japanese society where these impacts manifest most clearly. In the context of Japan, it is likely the young generation in their 20s and 30s. This section of the total demography is much more likely to be the primary drivers of this movement due to their proximity to the collective memory of Japan's postwar economic ascendancy, while at the same time growing up in environments that embrace traditional or patriotic values yet unexposed to the terror of war. (Wallace, 2016) This has resulted in high levels of engagement with the nationalist narratives of Japanese exceptionalism. Furthermore, lack of accurate historical education and the intentional weakening of negative impressions from the Second World war leads to a lack of acute historical consciousness. With this established, concepts such as patriotism or militarism are viewed as being aspirational and positive for the youths. This further purifies their idea of modern nationalism, and as a result is much more action-oriented as they truly believed in the glory it once brought. (Yang and Lim)

On the other hand, they are the major workforce in Japan with increasing burden due to the aging society. As previously established, the degradation of the economic landscape has led to less investment in human capital for new entrants to the workforce due to the need to cut costs, and there are very few opportunities to gain a high wage due to the valuation of seniority, resulting in a vicious cycle. This also coincides with the unwillingness of Japanese universities to invest in innovative technology and fields, resulting in an overall reduction of skills and wages for the younger generation as a whole. The point being made in this section is that the younger generation bore the brunt of the aforementioned structural failures. The preceding analysis indicates that in the context of Japan where aging continues to be a problem, the economic situation will only be much worse in the future if no effective action is taken. (Aoki, 2013)

A similarity can be seen in the generation, which is that they lived in a time when the situation rapidly changed in all aspects from prosperous to stagnation. Naturally, there is the tendency for them to gravitate toward the political movement that promises them to break the status quo. However, the distinction here is that the tangible economic burden is likely considered much more significant as a reason opposed to an idealistic appeal to national sovereignty considering the context of Japan now. The logic being that philosophically, the erosion of living standards exerts a

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more immediate pressure on the individual than the loss of national pride.

EVALUATION OF POLITICAL REFORM: REPERCUSSION VERSUS REWARD

The last comparison to be made here is the change in the political landscape itself, which is much more contentious in the context of nationalism. When analysing changes in government actions, proposed amendments of constitutional laws, and the outspoken desire for standing army certainly is much more bold and significant compared to economic policies at first glance. This is complemented by the extensive effort to shape the collective “Japanese Identity” through the emphasis on cultural education and the glorification of the imperial past. Ultimately, the core of the new strategy adopted by nationalist politicians is the cultural significance and emotional unification instead of traditional conservative campaigns that targeted the practical aspects like infrastructure. This is not to say policies like abenomics did not support the practical side, but is to say that the policies are uniquely centered around the concept of “independent development”, instead of development by itself. The differentiation here is key to their success, as it is not uncommon for policies that are similar to those proposed by the nationalists, but the “Japanese Pride” that is explicitly linked to the policies is both unique and compelling. This rhetoric resonated with the Japanese populace that has long been constrained by the postwar regime and its pacifist limitation, effectively garnering significant public support for a nationalist vision of collective resonance. (Wallace, 2016)

However, as mentioned in the economic section, some of the tangible repercussions of nationalist policies ultimately fed into the economic dissatisfaction without the population realising. The primary example of this dynamic is China, who deployed actions like economic sanctions, boycotts, and redirection of tourism from Japan to counteract Japanese actions. (Woodward, 2017) The practical impact of these political maneuvers ultimately manifested as further economic stagnation and contraction. This nuance in the outcome suggested that ultimately, the practical effect still manifested in the form of economic retaliation instead of further external pressure. This translation of political action into economic consequence suggests that although the changing political precedence regarding politics itself is arguably the strongest compared to all other aspects, the correlation between the process and the effect is inadequate to prove its overall effectiveness in persuading citizens. On the other hand, although the change in economic landscape is less revolutionary, the causal link between its economic drivers and their outcome is more direct and empirically verifiable.

In summary, both the scale and profundity of political conflicts as a mobilising force remains insufficiently resonant with the broader populace due to the nature of politics and limited tangible outcomes on those that are less engaged. On the other hand, though economic dissatisfaction amongst the population is less conspicuous, its universal generational scale ensures its primacy when compared to other aspects, not to mention the deep profundity and relevance it has to those that are impacted which leads to an overall decrease in productivity. Last, although the economic policies in the political landscape appear to be less significant compared to issues such as security, adequate justification can be provided as to why most systemic shocks across the political spectrum ultimately manifest in economic precarity. Consequently, the transition towards nationalism is driven more by the economical desire of the people rather than geopolitical threats or cultural ideologies.

CONCLUSION

In conclusion, the widespread dissatisfaction caused by economic stagnancy was significantly more instrumental in

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the resurgence of nationalism in modern Japan as opposed to prevailing political tensions. This is primarily due to the superior relevance and tangibility of economic impacts as they manifest across Japanese society. This is seen through the comparison of major political moves and incidents that are widely recognised as representative of the development, with recurring patterns that compares ideological significance to practical importance that illustrates the political landscape in modern Japan.

While some scholars caution against overemphasizing casualties, identifying the direct correlations between underlying drivers and their political manifestations remains essential to historical analysis. Investigating these nuances and making educated synthesis acts as a mirror to reflect the beliefs of present scholars towards the past. So to answer the question: To what extent was structural economic discontent the main cause of the resurgence of right-wing Japanese nationalism from the late 1990s to the modern era as compared to geopolitical anxieties? Both the pattern across the Japanese political and social landscape and the manifest of multicausal effects has proven the weight of prosperity over politics within the modern, practical scope of the contemporary scholarly debate.

Distinguishing between competing causalities is crucial in the present society due to the simple reason that this is the most accurate reflection of human nature and behaviour. By examining the causation, we are able to see the baseline of modern society, and the structure it orders upon. Historical and political research challenges the limit of human knowledge again and again by interrogating the social constructs under which we operate, only in the pursuit of a more comprehensive understanding of societal evolution.

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TO WHAT EXTENT DOES THE REDUCTION OF CROP STRAW BURNING CONTRIBUTE TO IMPROVING AIR QUALITY AND THE RESULTING RESIDENT HEALTH IN BAGUAZHOU, NANJING, CHINA, FROM 2010 TO 2025?

WRITTEN BY MUTONG (KATHERINE) WANG

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ABSTRACT

This study investigates the extent to which the reduction of open crop straw burning, following Jiangsu's 2009 ban, has contributed to improving air quality and resident health in Baguazhou, an agricultural island in Nanjing, China, over the period of 2010 to 2025. Drawing on official PM2.5 monitoring data, published literature, and a self-conducted survey of long-term Baguazhou residents as supplementary evidence, this paper found a very strong negative correlation with PM2.5 concentrations increasingly declining from 2010 to 2025 and suggests that reduced burning correlates with improved respiratory health in Baguazhou. The paper also reveals that most farmers stopped burning primarily due to external pressure rather than environmental awareness as well as the fact that stealth burning persists, suggesting that while the straw burning ban has been substantially effective in improving air quality and associated health outcomes, the sustainability of this progress depends on strengthening education and practical infrastructure alongside enforcement.

INTRODUCTION

To what extent does the reduction of crop straw burning contribute to improving air quality and the resulting resident health in Baguazhou, Nanjing, China, from 2010 to 2025?

LITERATURE REVIEW AND DEFINITION OF KEY TERMS

Existing studies have already shown that straw burning increases particulate matter and damages health. However, these studies usually focus on wider regional or national patterns in China or at a global scale, not on a specific community such as Baguazhou, Nanjing. This study therefore adds value to the academia by linking a local policy change to local air-quality data and residents' health outcomes over a fifteen year period.

Open biomass burning (OBB) refers to the combustion of vegetation in open areas, including forests, grasslands, shrublands, and agricultural fields. A major part of OBB comes from open crop straw burning (OCSB), which alone contributes to over 75% of total biomass burning emissions (Wen et al.), resulting in the heavy emissions of particulate matter (PM), especially PM2.5. This information is drawn from the journal *Environmental Science: Atmospheres*, which, according to SCImago Journal Rank (SJR) — a metric that measures journal prestige based on citation volume and the prestige of citing journals — is ranked in Q1 (Scimago), indicating that it is among the top 25% of journals in its category and is therefore a reliable source. According to the United States Environmental Protection Agency, the primary U.S. federal agency responsible for setting air quality standards PPM2.5 are fine inhalable particles with diameters that are generally 2.5 micrometers and smaller, that can get deep into human lungs and bloodstreams, causing respiratory health problems ranging from asthma attacks to cardiovascular disease and premature deaths (United States Environmental Protection Agency). Moreover, in a study published in a peer-reviewed article in *Frontiers in Environmental Science*, a reputable Q1 scientific journal (Scimago), OBB causes over 45% of PM2.5 concentrations around the globe (Mehmood et al.). China, as one of the world's largest agricultural producers, generates roughly 600 to 800 million tonnes of crop straw annually, which is about 20% of the total amount produced globally (Wen et al.). In China, widespread crop straw burning began in the mid-1980s as a cheap way to clear the fields after harvesting, and by the mid-1990s, it had spread across all major regions that produce grain. It was estimated that between 1997 and 2013, Chinese farmers burned about 2200 teragrams of crop stalks (Islam et al.), resulting in emissions of 1.036 million tons of PM2.5 per year, according to *Proceedings of the International Academy of Ecology and Environmental Sciences*, a peer-reviewed open-access journal focused on

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ecology and environmental sciences. During 2003-2005, satellite remote sensing detected an average of about 12800 burning sites per year across the nation, and Jiangsu Province had one of the highest counts. With about 2248 sites annually, the density of burning sites in Jiangsu reached 219.1 sites per 10000 square kilometers, more than 16 times the national average of 13 sites per 10000 km² during that period, according to World Bank. As a highly reputable international institution, data from the World Bank are generally reliable for macro-level comparisons, though the specific figures on burning sites from 2003–2005 may have limitations due to satellite detection constraints and spatial aggregation.

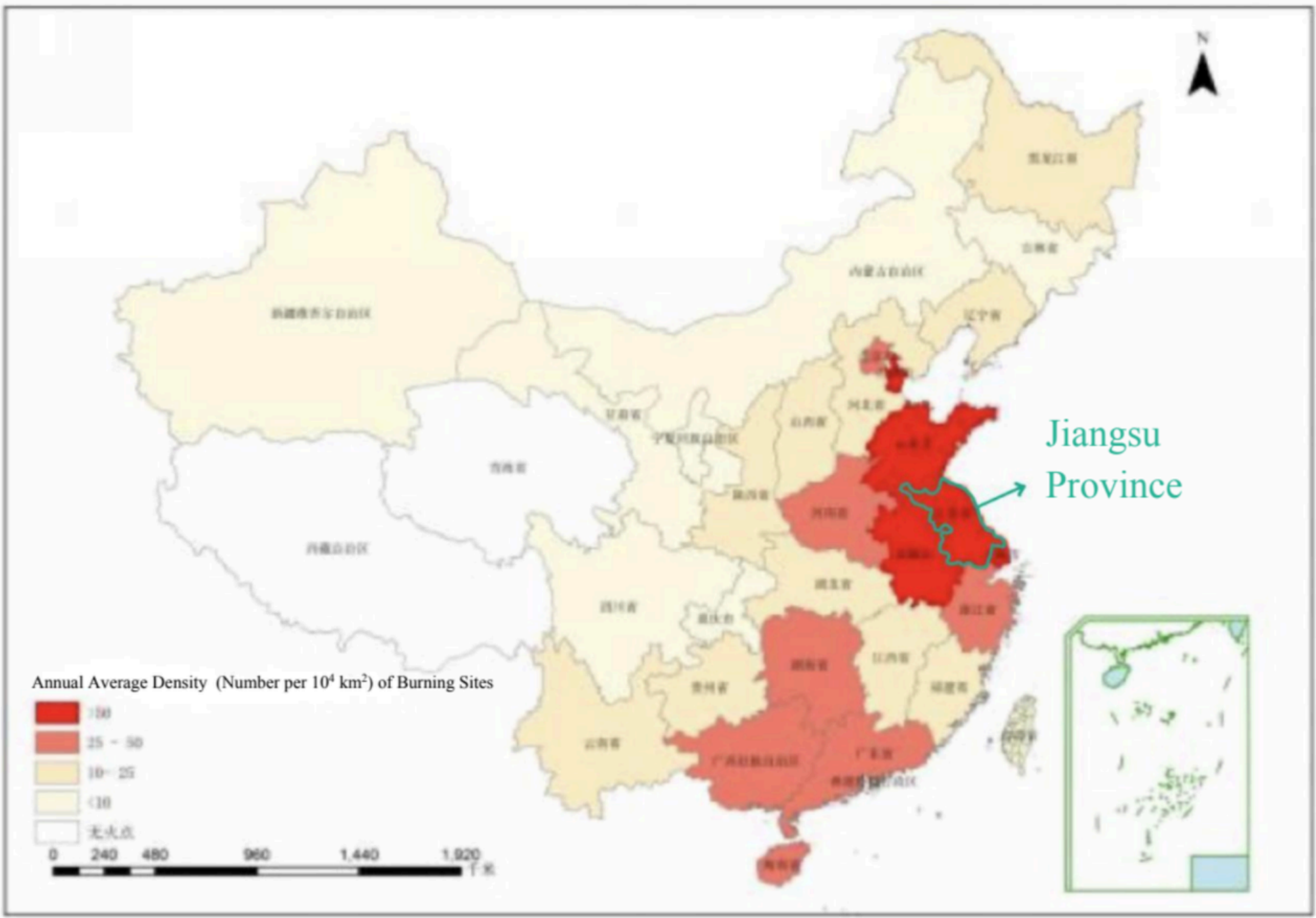


Figure 1. Annual Average Density (Number per 10⁴ km²) of Burning Sites in Various Chinese Provinces, Captured by MODIS Remote Sensing, 2003–2005 (World Bank)

However, this situation has changed. Starting in the late 1990s, the central government issued notices prohibiting straw burning near airports and highways, the gradually expanded the ban nationwide through stricter laws, leading to all-out bans in many provinces including Jiangsu. As a result, China achieved a sharp drop in emissions caused by crop straw burning and a rise in national straw-utilization rate, as stated in a study by Dongbei University of Finance and Economics and the University of Cambridge's Department of Land Economy, giving this piece of information high validity (Xu et al.).

GEOGRAPHICAL CONTEXT

Baguazhou Island, located in the northwest of Nanjing, Jiangsu Province, is an island covering 55.6 square kilometers between the Yangtze River and Jiajiang River. It is an agricultural area producing corn and rice during autumn harvest, from September to October (Baidu). Traditional straw burning created seasonal smog in Baguazhou (Xinhua News Agency), but national bans since 2009 (Standing Committee of the Eleventh Jiangsu Provincial People's Congress) promoted alternatives like composting or selling crop straw to factories that recycle crop straw, making it a suitable place to investigate how environmental regulation may affect both air quality and residents' health well-

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being. Although air pollution in Jiangsu has multiple sources, my study isolates crop straw burning by focusing on Baguazhou as an agricultural island where the main post-harvest residues are corn and rice straw, and where the 2009 ban directly targeted open burning of these residues.

HYPOTHESES

1. After the national ban in 2009, as more years passed, there is an increased reduction of crop straw burning in Baguazhou from 2010-2025, shown by the decline in regional annual average concentration of PM2.5.
2. Between 2010-2025, reduced burning correlates with improved health among residents, indicated by reduced premature mortality caused by cardiovascular disease (CVD) and chronic respiratory diseases.

METHODOLOGY

METHODS OF DATA COLLECTION

To investigate the relationship between crop straw burning reduction and residents’ well-being in Baguazhou, I used a mixed-method approach. I compiled official air quality reports from the People’s Government of Nanjing that provides annual average concentration of PM2.5 every year from 2010 to 2025, published studies on Nanjing citizen health, and other research from reliable sources such as universities or institutions to examine the trend in Baguazhou’s air quality after the 2009 ban and to judge whether the improvement in air quality was consistent with improvements in residents’ health. These data are highly reliable because they are either city-level official monitoring results or obtained by reputable institutions. For data used as supplementing evidence, I designed a survey and distributed it to Baguazhou residents who have lived on the island for more than 30 years. I used a snowball sampling method: I first sent the questionnaire to eligible residents I knew personally, such as my relatives living in Baguazhou, and then asked them to forward it to other eligible residents they knew. This approach was chosen because eligible residents in Baguazhou generally have low levels of formal education, making it difficult to find individuals who are both willing and able to complete a questionnaire. In such a situation, snowball sampling is the most efficient way to reach the target population (Hassan). A total of 45 valid responses were collected, 15 of them from farmers and 30 from non-farmers. The survey covered topics about past and current crop straw handling methods, change in respiratory health symptoms and life satisfaction before and after the burning ban, and so on. All responses were anonymous, and I informed participants that the data would only be used for my academic research. While such self-reported survey data are useful for illustrating residents’ lived experiences and for comparing with objective air quality trends, it must also be acknowledged that they are inherently subjective, vulnerable to recall bias, and based on a relatively small sample size, which may limit the generalizability and reliability of the findings. Therefore, these data are used only as supplementary and additional evidence, rather than as the primary support for my hypotheses.

METHODS OF DATA ANALYSIS

The method used is Spearman’s Rank Correlation Coefficient, which measures the degree of correlation between two data sets, suitable for determining whether a significant correlation exists between years passed since 2009 and PM 2.5 levels.

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Spearman correlation formula (Laerd Statistics):

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

ρ = Spearman's Rank Correlation Coefficient

d_i = difference in paired ranks

n = number of cases

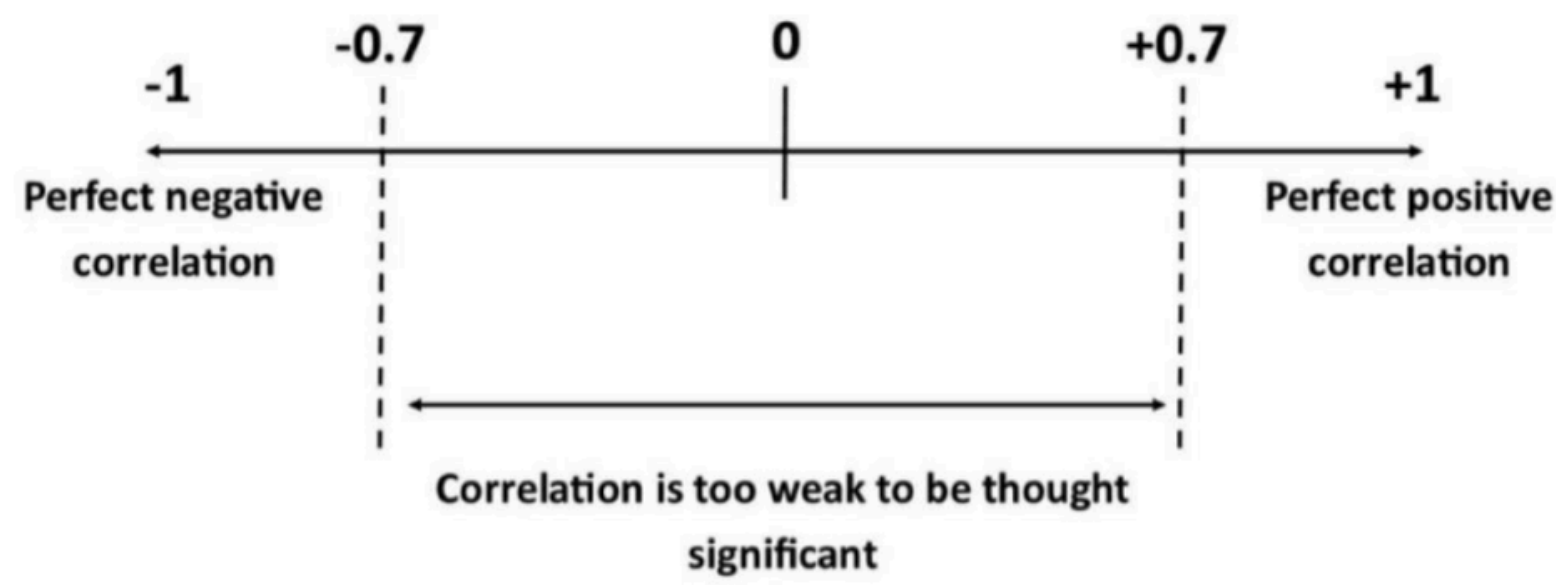


Figure 2. Spearman's Rank Correlation Coefficient Values (Royal Geographical Society)

DATA PRESENTATION AND ANALYSIS

HYPOTHESIS 1: AFTER THE NATIONAL BAN IN 2009, AS MORE YEARS PASSED, THERE IS AN INCREASED REDUCTION OF CROP STRAW BURNING IN BAGUAZHOU FROM 2010-2025, SHOWN BY THE DECLINE IN REGIONAL ANNUAL AVERAGE CONCENTRATION OF PM2.5.

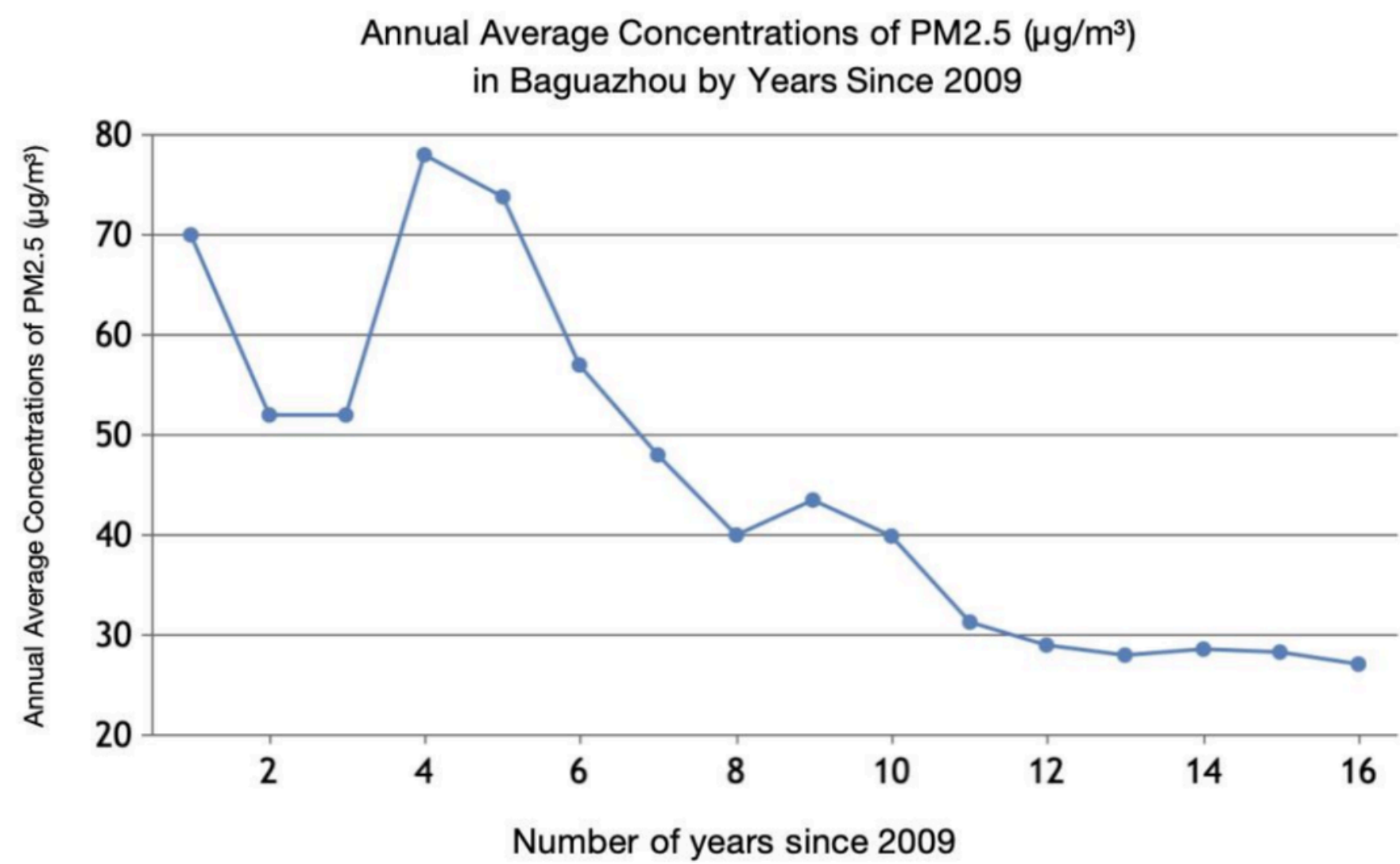


Figure 3. PM2.5 levels. Figure 3. PM2.5 Levels (µg/m3) in Baguazhou by Years Since 2009 (南京市人民政府 [The People's Government of Nanjing])

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This line graph is constructed with official data from The People's Government of Nanjing. From the line graph, it can be observed that the PM2.5 data for Baguazhou from 2010 (year 1 after the crop straw burning ban) to 2025 (year 16) shows a clear downward trend. In 2010, the concentration was 70 $\mu\text{g}/\text{m}^3$. It fluctuated in the first few years after the ban, peaking at 78 $\mu\text{g}/\text{m}^3$ in 2013 and 73.8 $\mu\text{g}/\text{m}^3$ in 2014, which can imply that the ban took time to be fully effective. During earlier years, it is likely that law enforcement was not sufficient and sustainable alternatives to burning crop straw had not been widely developed, so a large part of farmers still burned crop straw out of habit. However, after 2014, the PM2.5 concentrations started to drop at a steady pace. In 2025, 16 years after the ban, the annual average PM2.5 concentration reduced to as low as 27.1 $\mu\text{g}/\text{m}^3$, fallen by more than 60% compared to the value in 2010. This shows that as more years passed, the effect of the ban became increasingly visible. This pattern can be explained by the fact that the 2009 Jiangsu straw-burning ban directly targeted open crop straw burning, which is a major source of PM2.5 and other particulate matter that pollutes the air, as stated in Section 1.2 (Wen et al.). Over the following years, enforcement became stricter, with more patrols, heavier fines, and clearer guidance on alternative straw disposal methods, as officially stated in reports published by The People's Government of Qixia District, Nanjing (南京栖霞区人民政府 [The People's Government of Qixia District, Nanjing]). As a result, fewer farmers continued open burning, so the reduction in burning is a plausible explanation for the decline in PM2.5 rather than a coincidence. This matches well with the observed change in farmers' behavior presented by my survey results. Before the ban, 20% of the farmers burned crop straw directly in the fields or by the road, and 60% took it home to use as fuel, much of which was likely burned as well. Only 20% of farmers chose to sell straw to recycling factories.

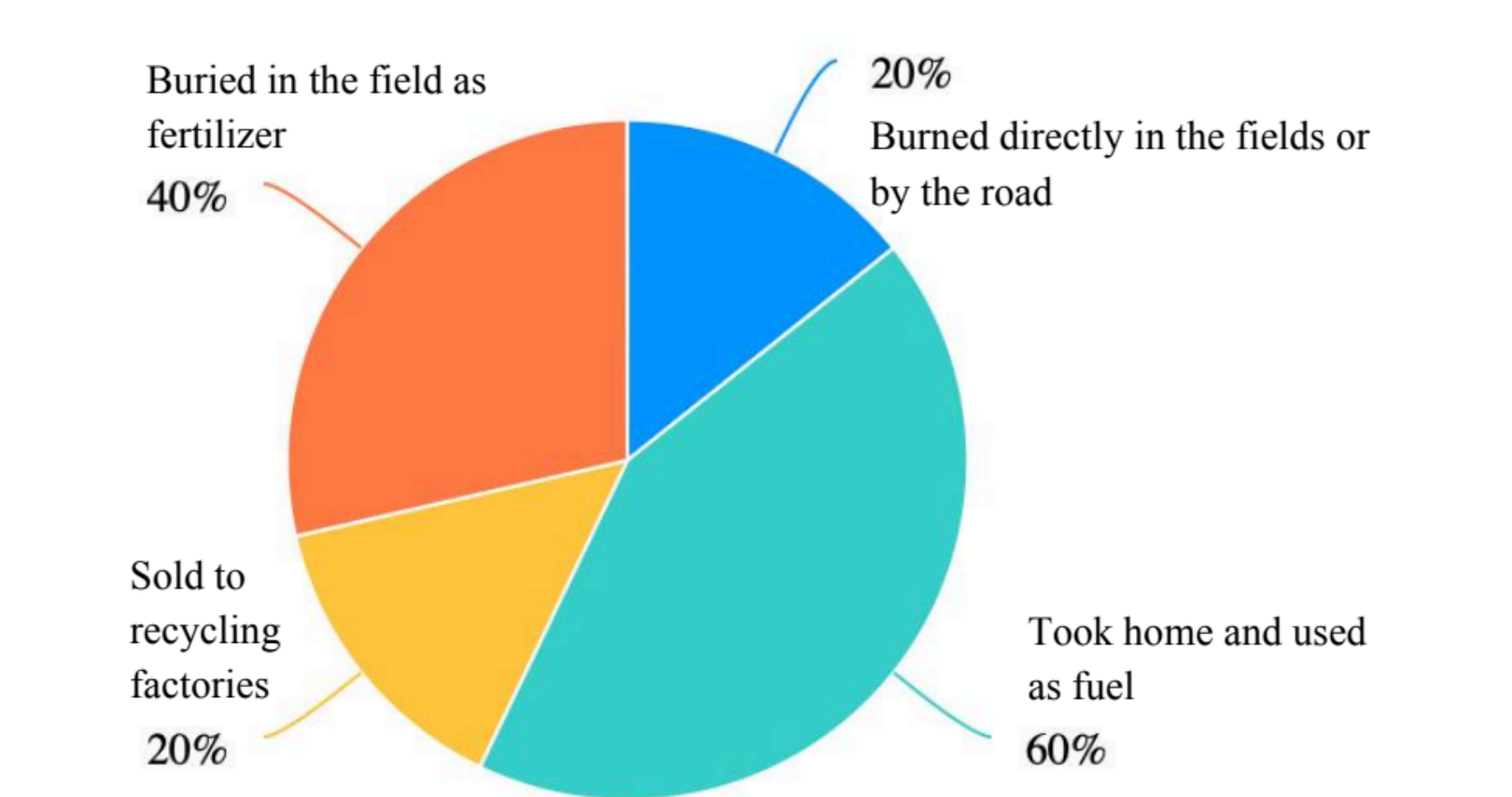


Figure 3.1 Proportion of Farmers Reporting Each Crop Straw Disposal Method Before 2009 Ban

After the ban, none of the farmers responded that they burn crop straw openly in the fields, and only one person (6.67%) still piled straw by the road to burn. The proportion of farmers burning crop straw as fuel dropped to 26.67%, and the proportion of farmers selling straw to factories rose to 46.7%, possibly because the straw recycling method is more developed and accepted. This is objectively corroborated by a Xinhua News report on Baguazhou, which describes a government-supported “enterprise plus agent” recycling system, where a local factory purchases over 40,000 tonnes of crop straw from farmers annually, farmers receive a subsidised price of 40 yuan per truckload, and one interviewed farmer earns over 50,000 yuan per year from selling straw, demonstrating that economic incentives and improved collection services have made selling crop straw a realistic and profitable alternative to burning (Xinhua News Agency). However, it must be acknowledged that survey respondents may be reluctant to

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admit illegal behavior even under conditions of anonymity, as open burning has been prohibited by law since 2009. Therefore, the reported post-ban burning figures likely underestimate the true extent of continued burning, which is a major limitation of the self-reported data I collected on sensitive topics. Consequently, while the survey suggests a substantial reduction in open burning, the actual persistence of straw burning may be higher than the numbers indicate.

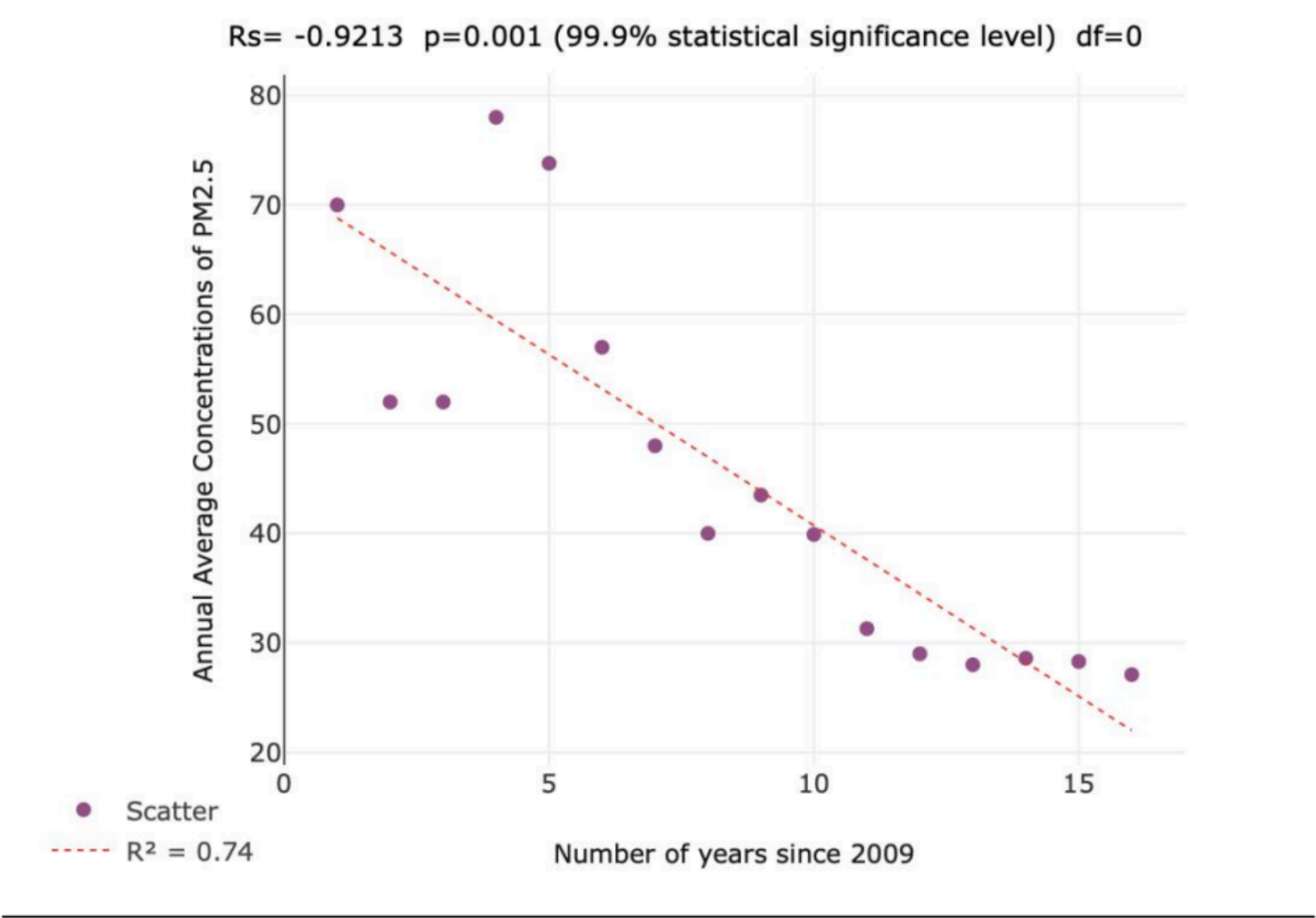


Figure 3.2 Scatter Graph for Spearman Correlation (Barcelona Field Studies Centre)

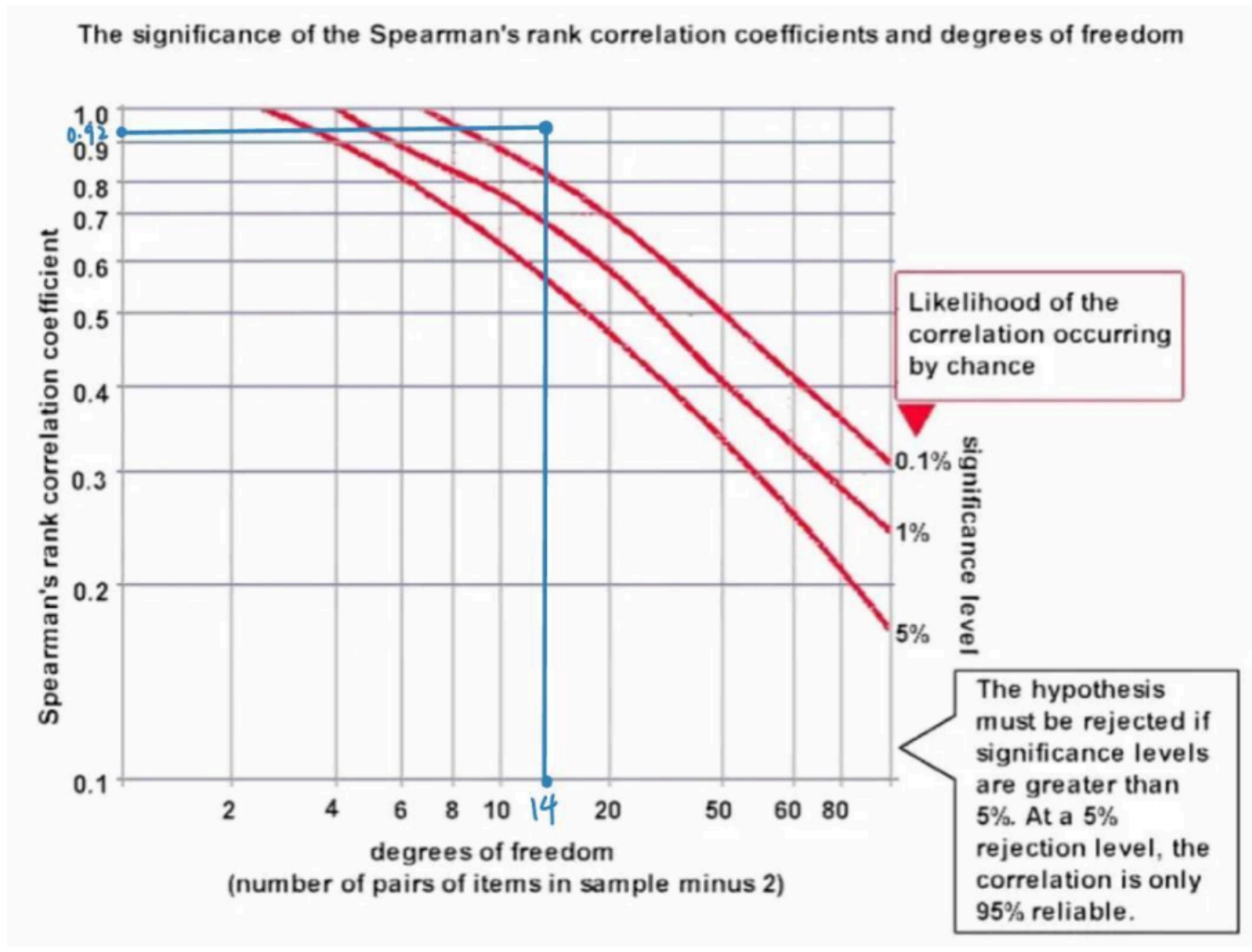


Figure 3.3 Graph of significance level for Spearman Correlation (Barcelona Field Studies Centre)

When applying the Spearman's Rank Correlation Coefficient, a Spearman coefficient of -0.92 is reached, indicating a very strong negative correlation. The significance level shows that this correlation's likelihood of occurring by chance is lower than 0.1%, meaning the correlation must be accepted. This means that as the number of years after 2009

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increases, PM2.5 levels in Baguazhou fall. To understand why this happened, my survey asked farmers who no longer burn straw about their main reason for stopping. 40% said it was because the government bans burning and they fear being fined. Another 40% said they stopped because they know burning pollutes the air and is bad for health. Notably, one person (6.67%) admitted that they still burn sometimes. This implies that while the ban has been effective in reducing straw burning and improving air quality, more than half of the farmers' motivation to stop came from external pressure, mainly the fear of getting caught and punished, rather than a deep internal belief that burning is harmful. Only 40% reported health and environmental awareness as their reason for stopping burning crop straw. This suggests that if enforcement weakens, many farmers are likely to return to burning. This point is further supported in an article published by Chinese Academy of Agricultural Sciences, where Minchun Zhang, the Dean of Changsha University of Science and Technology Graduate School and Law School, stated that in the regions of China where straw burning bans are enforced, policy implementation tends to focus narrowly on achieving compliance outcomes, while public education and awareness campaigns remain insufficient. Consequently, many farmers hold resistant attitudes toward the bans (Chinese Academy of Agricultural Sciences). Looking at it from a farmer's perspective, it is understandable. For generations, burning was the easiest and cheapest way to clear fields after harvest (法制日报 [Legal Daily]). Changing a habit takes time, and forcing a new rule without fully changing people's understanding often leads to compliance only on the surface. This implies that when governments design environmental policies like crop straw burning bans, they often focus on enforcement, such as patrols and fines, but the importance of education and building awareness are often forgotten. If policymakers want a lasting solution, investment in teaching stakeholders the reason behind the policy is crucial. In conclusion, the reduction in crop straw burning did increasingly lower PM2.5 levels over time, and Hypothesis 1 is supported. Yet research also reveals that most farmers stopped burning because of external pressure instead of environmental awareness, which means that while air quality has improved, a stronger focus on farmers' education is needed to keep the progress fully sustainable.

HYPOTHESIS 2: BETWEEN 2010-2025, REDUCED BURNING CORRELATES WITH IMPROVED HEALTH AMONG RESIDENTS, INDICATED BY REDUCED PREMATURE MORTALITY CAUSED BY CARDIOVASCULAR DISEASE (CVD) AND CHRONIC RESPIRATORY DISEASES.

Crop straw burning releases fine particulate matter into the air, and this pollution can irritate the respiratory system and damage exposed populations, as stated in Section 1.2. Therefore, when crop straw burning decreases, PM2.5 exposure also decreases, which reduces cough, throat discomfort, nasal irritation, and other respiratory symptoms. This reasoning is supported by evidence based on long-term studies published on BMC Public Health, a peer-reviewed open-access journal (Yang et al.). The journal was placed in the top 25% (Q1) of journals in its subject category by SJR ranking and has an H-Index of 225 (Scimago), showing that it is not only currently well ranked but is also frequently cited, making it a credible and prestigious source within public health. One study published on BMC Public Health investigated effects of air pollution on influenza-like illness in Nanjing, and showed that higher PM2.5 exposure leads to more cases of influenza-like respiratory illness (Huang et al.), while another study extracted data from the Death Information Registration and Management System of Chinese Center for Disease Control and Prevention and found that premature mortality from CVD and chronic respiratory diseases in Nanjing decreased with the average annual percent changes (AAPC) at -5.9% and -1.6% respectively, between 2007 and 2018. These findings support the view that improved air quality can contribute to improved residents' physical health. My survey results provide supplementary evidence. When asked to compare their autumn respiratory symptoms including coughing, throat discomfort and nasal discomfort between the pre-ban period (15 years ago) and now, using a 10-

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point scale where 1 = much worse and 10 = much better, most respondents reported that their symptoms had improved in recent years, with average scores of 7.33 and 7.38 out of 10. Although these self-reported data are less reliable than official statistics or medical studies and have the limitation that individuals may hold different personal standards for what constitutes “much better,” making the scores not directly comparable across respondents and potentially introducing memory bias or varying health expectations, they are still consistent with the objective trend and show that residents themselves noticed the improvement.

CONCLUSION

This study aimed to answer: to what extent does the reduction of crop straw burning improve air quality and residents’ well-being in Baguazhou from 2010 to 2025? The answer is: to a substantial extent, though limitations remain. To start with, air quality improved significantly after the 2009 ban of crop straw burning. PM2.5 levels dropped from 70 $\mu\text{g}/\text{m}^3$ in 2010 to 27.1 $\mu\text{g}/\text{m}^3$ in 2025. In addition, a very strong negative Spearman correlation of -0.92 confirms that this decline is closely connected to the passage of time since the ban, during which farmers gradually stopped burning straw. As a direct consequence of cleaner air, residents’ respiratory health improved, indicated by published studies in Nanjing that show a reduction in premature mortality from CVD and chronic respiratory diseases over the same period. Taken together, these results make it reasonable to conclude that the reduction in straw burning was a major contributor to better air quality, which in turn led to measurable improvements in local health outcomes. However, the transition is not perfect. Yangtze Evening Post, which is Jiangsu Province’s official newspaper and the evening newspaper with largest national circulation in China, reported stealth burning still exists in Baguazhou (Yangtze Evening Post). Moreover, multiple official announcements from the Qixia District government repeatedly mention the need for inspection and supervision to prevent unauthorized straw burning in Baguazhou (南京栖霞区人民政府 [The People's Government of Qixia District, Nanjing]). This shows that despite policy efforts, illegal burning persists and requires ongoing enforcement. My survey adds supplementary information: 57% of residents have seen or heard of stealth burning in Baguazhou, and the main reasons are farmers believe that burning crop straw is more convenient than bringing them to recycling companies, as well as limited accessibility of collection services. This implies that while the policy has been largely successful, leftover problems remain. The effectiveness of a ban depends not only on laws and penalties, but also on practical support and on changing farmers’ habits through education.



Figure 4. Baguazhou Farmer Burning Crop Straw in the Field (Author’s Personal Collection)

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According to the results, it is revealed that most farmers stopped burning not because of a deep understanding of the environmental or health risks, but because they feared fines. This offers implications for future policy design in other agricultural regions of China and beyond: compliance driven by force can initiate change, but if enforcement weakens or services offering alternatives remain inadequate, old practices are likely to return. The research therefore suggests that to achieve reduction of crop straw burning and environmental protection, having legal bans alone is insufficient, and long-term success requires investment in education to build environmental awareness, as well as in infrastructure, such as more accessible and reliable straw collection systems.

EVALUATION

STRENGTHS

The study has several strengths. All survey responses were anonymous, and I clearly stated that the data would be used only for academic research, which encouraged honesty. By selecting only long-term residents (over 30 years on the island), I ensured that respondents could remember the situation before the ban. The combination of primary survey data with secondary PM2.5 data allowed me to link subjective well-being to objective air quality trends.

LIMITATIONS AND IMPROVEMENTS

1. The sample is small (45 people) and not random – while approximately 50% of the respondents were residents I randomly selected to interview during my visit to Baguazhou, 50% were my relatives or acquaintances, which could introduce bias for that participants might not be absolutely honest when self-reporting information. A larger, randomly selected sample would give more reliable results. To overcome this in my paper, I used the survey only as additional evidence to supplement my primary evidence, which is published literature from reliable sources like renowned journals and institutions. I also explicitly acknowledged the bias and limitations of my survey data every time it was mentioned, to minimize the inaccuracies that might be caused.
2. The official PM2.5 data published by the Nanjing Government comes from a single monitoring point, so it is difficult to verify whether that point perfectly represents the whole island. Also, the data do not separate straw-burning pollution from industrial or traffic pollution. To improve this, future studies could use portable PM2.5 monitors placed near fields during harvest seasons. In the broader scheme of my paper, I addressed this by focusing on the long-term downward trend rather than specific values, and by using Spearman's rank correlation which only requires consistent relative measurements. This helps demonstrate the effect of the ban despite the lack of source-specific data.
3. Other things also changed in Baguazhou between 2010 and 2025 besides reduction of crop straw burning. New roads, cleaner cooking fuels, and other factors are all able to affect local air quality and well-being. Because my study did not control for these, I cannot claim that straw burning reduction is the only cause of the improvements, only that it is a very likely major contributor. To mitigate the influence of this limitation, I explicitly acknowledged these confounding factors in my discussion and did not claim an exclusive causation.

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APPENDIX

Raw data and responses obtained from the survey for Baguazhou residents:

关于八卦洲秸秆禁烧对居民影响的调查

第 1 题 您的身份是? [单选题]

选项	小计	比例
A. 农民	15	33.33%
B. 非农民（在八卦洲居住，但不种地）	30	66.67%
本题有效填写人次	45	

第 2 题 大约 20 年前（2005-2010，没有禁烧秸秆时），您家地里的秸秆主要怎么处理? [多选题]

选项	小计	比例
A. 在田里或路边烧掉	3	20%
C. 拿回家当柴火	9	60%
D. 卖给工厂或收秸秆的人	3	20%
E. 埋到土里当肥料	6	40%
本题有效填写人次	15	

第 3 题 最近 5-10 年（秸秆禁烧后），您家地里的秸秆主要怎么处理?（单选） [多选题]

选项	小计	比例
A. 直接在田里烧掉	0	0%
B. 堆在田边/路边烧掉	1	6.67%
C. 拿回家当柴火	4	26.67%
D. 卖给工厂或收秸秆的人	5	33.33%
E. 埋到土里当肥料	8	53.33%
F. 其他（请注明:）	2	13.33%
本题有效填写人次	15	

第 4 题 如果您现在不再烧秸秆了，最主要的原因是什么? [单选题]

选项	小计	比例
A. 政府不让烧，怕被罚款	6	40%
B. 卖秸秆可以增加收入	0	0%
C. 烧秸秆污染空气，对身体不好	6	40%
D. 村里有人来收，方便处理	2	13.33%
E. 邻居都不烧了，我也不好意思烧	0	0%
F. 我还是会烧一些	1	6.67%
G. 其他原因（请注明:）	0	0%
本题有效填写人次	15	

第 5 题 您是否听说过或了解有人还在偷偷烧秸秆? [单选题]

选项	小计	比例
A. 经常看到/听说	3	21.43%
B. 偶尔看到/听说	5	35.71%
C. 很少看到/听说	2	14.29%
D. 完全没有	4	28.57%
本题有效填写人次	14	

第 6 题 如果有人偷偷烧秸秆，您觉得主要原因有什么? [多选题]

选项	小计	比例
A. 没有人来收秸秆，没地方处理	8	57.14%
B. 卖秸秆的钱太少，不值得	3	21.43%
C. 觉得烧了方便省事	10	71.43%
D. 觉得偷偷烧没人管	1	7.14%
E. 不清楚	2	14.29%
F. 其他（请注明:）	0	0%
本题有效填写人次	14	

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第 7 题 与 20 年前（秸秆禁烧前）相比，您觉得自己秋天时的咳嗽、嗓子不舒服情况有什么变化？ [\[量表题\]](#)

本题平均分：7.33

选项	小计	比例
明显变严重了	4	8.89%
1	0	0%
2	1	2.22%
3	1	2.22%
4	3	6.67%
5	6	13.33%
6	0	0%
7	1	2.22%
8	5	11.11%
9	4	8.89%
明显好转了	20	44.44%
本题有效填写人次	45	

第 8 题 与 20 年前（秸秆禁烧前）相比，您觉得自己秋天时的鼻子不舒服/打喷嚏情况有什么变化？ [\[量表题\]](#)

本题平均分：7.38

选项	小计	比例
明显变严重了	5	11.11%
1	0	0%
2	1	2.22%
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TAKE YOUR MEDS: REDUCING MOTIVATIONAL BARRIERS FOR NON-ADHERENCE HYPERTENSIVE PATIENTS THROUGH INCREMENTAL MINDSETS

WRITTEN BY YUEQI GOU

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ABSTRACT

Non-adherence within hypertensive patient populations is a prevalent but complex issue that stems from both constitutional failures and personal decisions. There are three most prominent pillars that contribute to this phenomenon: communicational, socioeconomic and motivational factors. This paper explores the possible application of a growth mindset in resolving motivational issues by increasing the perceived value of treatment and fostering perseverance when faced with medical roadblocks. However, other contributors to non-adherence, such as socioeconomic and communicational barriers, limit the effect of growth mindset application.

INTRODUCTION

Hypertension is a highly prevalent medical problem that occurs when blood pressure in the vessels is consistently elevated. WHO estimates that there were around 1.4 billion people living with hypertension in 2024.¹ Although frequently asymptomatic for most individuals, untreated hypertension significantly increases the risk of stroke, heart attacks, chronic kidney disease, and dementia. Treatment of hypertension relies on lifelong medication and potential lifestyle changes, requiring dedication and consistency in treatment plans and schedules.² However, a key issue emerges: patient non-adherence to therapy regimens.

Non-adherence refers to an observed phenomenon in which patients do not follow prescribed treatment recommendations from their healthcare provider. Individuals who do not follow medication plans, either by not initiating medication or by discontinuing early, often see the poorest clinical outcomes and may result in severe side effects.³ Non-adherence to medicine is a global issue and 43%–65.5% of noncompliant patients are hypertensive patients who do not adhere strictly to their prescribed medications as recommended,⁴ highlighting a widespread issue not only as a clinical failure resulting in preventable cardiovascular events but also as a massive economic burden on healthcare infrastructure.⁵ In order to resolve this issue, researchers have grouped barriers to maintaining medication regimens into three key sectors: socioeconomic, communicational, and motivational. It is important to acknowledge the root cause of noncompliance in order to effectively examine how noncompliance can be reduced.

While studies have focused extensively on key factors contributing to, and methods to identify, non-adherence, little research has been conducted to examine the effectiveness of mindset changes on improving adherence. Despite its usual application in child psychology and education, current literature suggests that exhibiting a growth mindset can greatly improve adherence to therapy in a medical setting. According to Implicit Theories of Intelligence, mindsets (referred to as implicit theories) are responsible for both motivational and cognitive frameworks that influence the individuals' views and responses to learning engagement and achievement.^{6,7} They define an individual with a growth mindset (referred to as an incremental theorist) as focusing on achieving mastery in subject areas and using performance outcomes as feedback opportunities to self-reflect. In contrast, fixed-mindset individuals (referred to as entity theorists) believe intelligence to be a static, immutable quality, treating performance opportunities as a definitive statement of their ability. Entity theorists give up prematurely and lose motivation if results do not align with expectations, rather than practising reflection and improving upon faults.^{8,9}

Another key factor that predicts student learning and achievement is the possession of intrinsic motivation, defined as a force that encourages behaviour and self-determination due to inherent enjoyment.¹⁰ Its counterpart, extrinsic motivation, is the driving force to achieve something for the reward offered for completing the task.¹¹ Of the two,

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intrinsic motivation has been deemed to be responsible for lifelong learning by the general academia,¹⁰ as it encourages perseverance despite external setbacks.

The application of incremental theories and encouragement of intrinsic motivation improves patient adherence in a healthcare setting by resolving the motivation barrier. As identified earlier, patient motivation is a significant contributor to non-adherence, as they might underestimate the importance of completing treatment plans. Hypertension is also classified as an oligosymptomatic disorder, which is a disorder that generally shows mild symptoms¹²; therefore, long-lasting therapy regimens that may become repetitive or seem unrewarding over time. This view likely stems from the lack of external validation associated with curative results such as a major decrease in symptoms and discomfort; hence, patients become dissuaded. Therefore, this paper seeks to investigate the extent to which adopting an incremental mindset can improve therapy adherence in patients with severe hypertension by increasing the perceived value of prescribed medical therapy. However, it is also important to acknowledge that patient non-adherence stems from many holistic factors that may become barriers to following therapy regimens rather than a conscious choice to stop taking medication.

Throughout this paper, the root causes of non-adherence are going to be thoroughly examined in the context of hypertensive patients in order to holistically understand how it could be effectively reduced in patient populations. Growth mindsets and intrinsic motivation are then going to be introduced and reviewed in their original context of education, leading to evaluations on the potential application of these interventions in decreasing noncompliance in the context of medicine. Factors that limit the effectiveness of growth mindset application will also be thoroughly identified and explained.

MAIN CAUSES OF NON-ADHERENCE

Understanding the causes of non-adherence is fundamental to analysing the effectiveness of growth mindset therapy on improving medication adherence. The oligosymptomatic nature of this disease requires intrinsic dedication in both lifestyle and medication compliance for long periods of time, although treatment may appear unsatisfactory.¹³

A current method to explain causes of patient non-adherence can be attributed to the introduction of the Health Belief Model (HBM) by social psychologists in the 1950s. Through multi-faceted assessments in areas of perceptions such as predisposition, significance, gravity, benefits and obstacles to a health behaviour, the HBM can accurately examine and predict individual behaviours relating to prevention and control of disease and symptoms. Rosenstock et al. synthesised that treatment compliance is heavily reliant on perceived susceptibility, perceived significance, perceived obstacles, perceived benefits and cues to action (Figure 1.). At its core, the three main pillars of non-adherence can be summarised as motivation, belief of vulnerability, and trust in recommendations.

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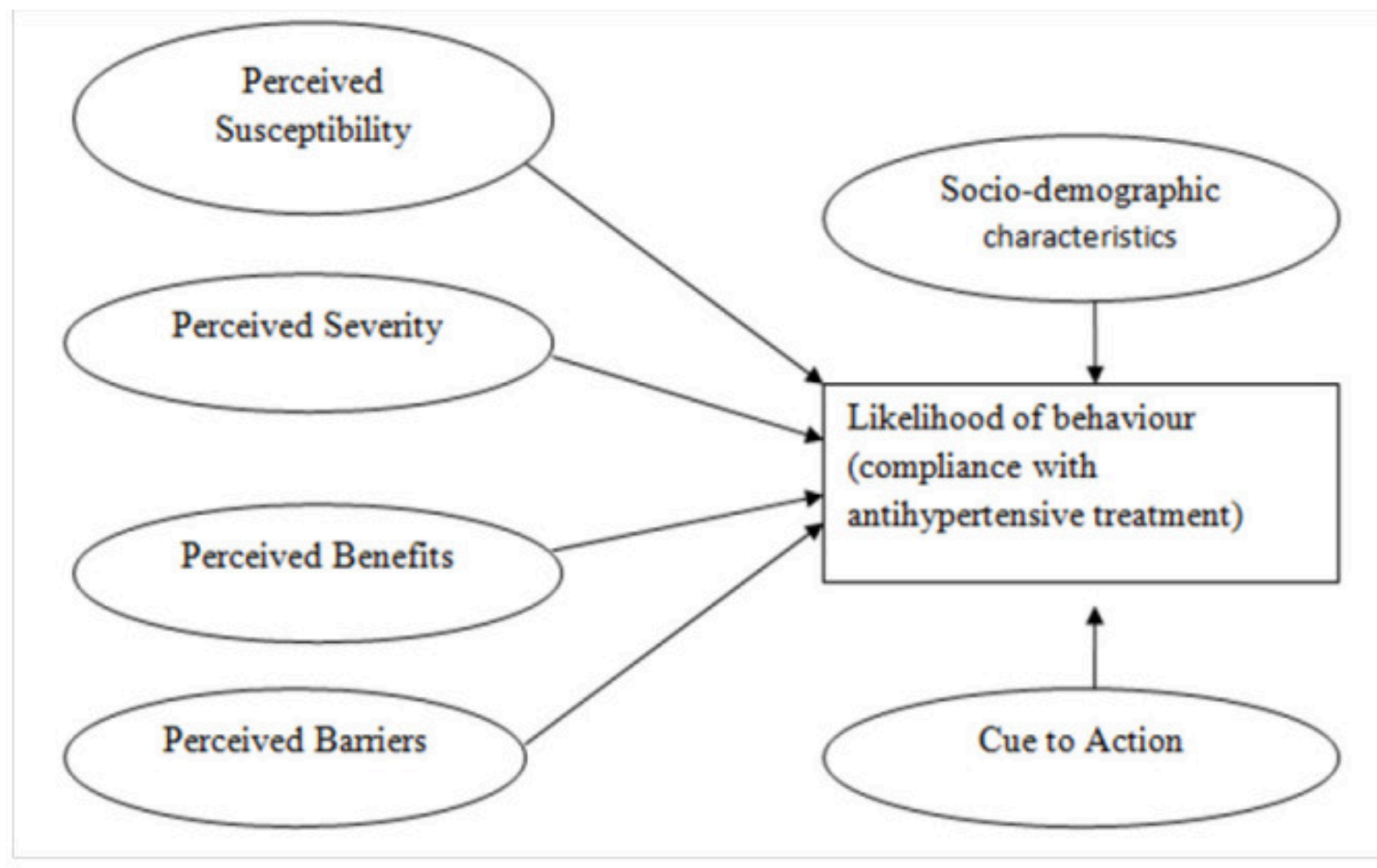


Figure 1. Demonstrating the relationship between key variables of HBM and treatment compliance behaviour.⁵

While it was originally used to assess patient motives behind an individual's participation in health programmes, a study by Joho⁵ (2021) examined how key factors of HBM can influence treatment compliance within hypertensive patients in Dar es Salaam, Tanzania. In her study, a cross-dimensional approach was used to analyse data from 3 regional hospitals in Dar es Salaam. Random sampling was used to select hypertensive patients which met the inclusion criteria, which required patients to be at least 18 years and above with an existing diagnosis of hypertension for at least one month (therefore have been taking anti-hypertensive treatment for at least one month prior to the beginning of the study), regardless of whether they had a coexisting medical condition. All participants have consented to participate in the study.

Data were collected in areas of perceived significance, benefit, barriers, lifestyle modifications and adherence to therapy regimens in the form of survey questions, in which participants responded with decreasing frequency. For example, a question on treatment compliance, asking “How often do you forget to take your medicine?”, is scored from 1-4 on a 4-point Likert scale: “1” every day, “2” frequently, “3” rarely or “4” never. However, this meant that the highest score may not be entirely indicative of non-compliance. This is illustrated by the question “How often do you engage in physical exercise?” for which a higher score of “4” indicates better compliance as opposed to “1”. In these cases, scores were reversed for more accurate scoring.

The results from the study demonstrate a positive relationship between perceived severity, perceived benefit and treatment compliance. Participants with high perceived severity (57.4% compliant) and benefit (66.7% compliant) were more likely to follow treatment regimens than participants without (53.7% and 44.9% compliant, respectively). As expected, a negative relationship was shown between perceived barriers and treatment compliance. A lower perceived barrier predicted higher treatment compliance, at 77%, compared to those with a high perceived barrier, who had only 29.5% compliance. The study also revealed that compliance can be heavily influenced by socio-demographical factors such as sex. Female participants (63%) reported a higher percentage of compliance compared

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to male participants (46%). The other socio-demographical factors such as marital status, age, education, and occupation were also explored but did not appear statistically significant.

These findings lead to the conclusion that perceived susceptibility, benefit, and barriers are all key contributing factors to patient non-adherence in treatment regimens. Although some socio-demographical factors such as marital status, age, education and employment were not statistically significant within this experiment, they may be important determinants of compliance for individuals living outside of Dar es Salaam, Tanzania.

A study¹⁴ on cost-related medication noncompliance (CRN) using data from the National Health Interview Survey found that medication unaffordability was the strongest predictor of CRN, showing consistent findings along with another study by McHorney¹⁵ in 2008 and by McHorney et al.¹⁶ in 2010. These results suggest that unaffordability of healthcare may not be a prevalent issue in Dar es Salaam, Tanzania; hence, differences in employment statuses were not statistically significant. However, as demonstrated by the conclusion from the above studies, socioeconomic status is an important determinant of adherence for the rest of the world.

Another model used to demonstrate and explain factors of non-adherence is the Venn diagram model (Figure 2.) constructed by Baroletti¹⁷ et al. in 2010. This model highlights the three pillars of treatment noncompliance as socioeconomic, motivational and communicational barriers.

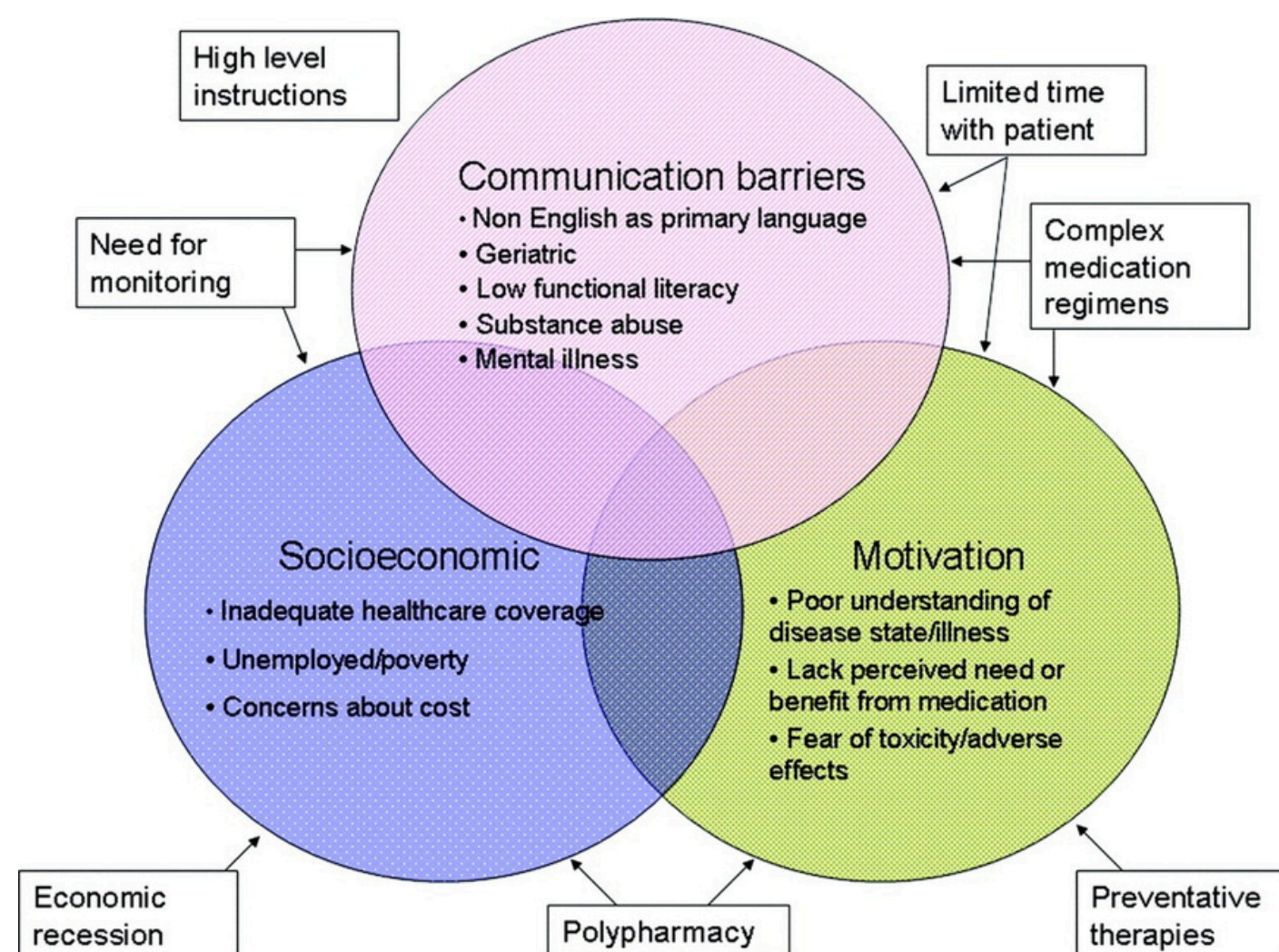


Figure 2. Demonstrating the factors contributing to non-adherence and overlaps¹⁷

As illustrated by Figure 2, socioeconomic barriers are determined by the financial background of individuals, specifically an individual's inability to afford medication. Consistent with previous studies on CRN¹⁴, patients who struggle with high costs of chronic treatment may involuntarily choose to drop medication plans, despite having full awareness of the consequences if they choose to do so.¹⁶ Chronic conditions such as hypertension further amplify this issue and in addition to inadequate healthcare coverage, long-term treatment medication costs can seem like a financially insurmountable obstacle.

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Communication barriers can be attributed to the difficulty in understanding treatment plans, often resulting in mistrust, misunderstanding, or misconception. The many sub-factors that contribute to communication issues between provider and patient can be organised into two main groups: Difficulty understanding instructions or low levels of trust with the provider. A patient may face obstacles with understanding instructions if they are low in functional fluency of the language, are of elder age or face mental illness or substance abuse.¹² Mistrust may also occur if patients face discriminatory behaviours enacted by providers, with 53% of patients reporting non-adherence to physician recommendations.¹⁸

Consistent with the conclusion drawn from the HBM study,⁵ motivation to follow treatment regimens is a crucial factor in determining behaviours of non-adherence. As identified earlier, the 4 main determinants are perceived susceptibility, perceived significance, perceived obstacles and perceived benefits of medication. In this model, perceived susceptibility and significance can be classified as having a poor understanding of the disease, while fear of adverse effects could count as an obstacle to compliance.

It is equally important to acknowledge the overlaps between the three main pillars in order to holistically examine the causes of non-adherence. As illustrated by Figure 2, a high need for monitoring may cause both socioeconomic and communicational pressure on the patient. Difficulty in understanding treatment plans could cause unintentional non-adherence due to a lack of perceived need for self-monitoring and language barriers cause difficulty rescheduling or understanding check-up timings.¹⁹

Medical appointments could also become financially taxing on individuals if there is inadequate healthcare coverage. Limited time spent with a patient can contribute to both communicational and motivational barriers, as there may not be enough time to fully explain the importance of therapy regimens or how to take them. This can cause a lack of perceived significance of medication and poor understanding of disease progression. The issue is amplified when medication regimens are complex and difficult to follow, causing unintentional non-adherence if patients do not understand how to take medication appropriately. Confusion caused by unclear instructions may contribute to a poor understanding of the illness. Polypharmacy, defined as regularly using five or more medications in the same time period,²⁰ is an issue that may intensify both socioeconomic and motivation barriers within patient communities. This is due to a decrease in the perceived need for taking extra medication and the accumulation of such drugs may be financially taxing overtime, especially if they are all used to manage chronic conditions.

While models thoroughly explain all holistic factors that contribute to medication noncompliance for hypertensive patient communities, all reasons can be summarised into three main pillars: motivational, socioeconomic, and communicational barriers. The majority of non-adherence stems from a lack of perceived need and benefit for taking medication; therefore, it is crucial to explore affordable methods to mitigate these motivational obstacles. A potential procedure could be the adoption of an incremental mindset by patients to enhance their intrinsic motivation and perceived value of taking medication.

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INCREMENTAL THEORY AND INTRINSIC MOTIVATION

Understanding the theory behind incremental theories and intrinsic motivation is crucial to analysing its potential application in the healthcare field as a possible method to mitigate motivational issues. Incremental theory, commonly known as growth mindset theory, has been a common focus of research regarding topics in child psychology and learning.

Students who hold incremental mindsets do not view mistakes as definitive judgements of their ability,^{6,7} treating learning as an opportunity for growth rather than a testament to one's intelligence. This approach increases the perceived value of the learning process instead of the result, shifting the focus from performance to mastery. Many positive correlations have been found between adopting an incremental approach and higher academic achievement,^{21,22} likely stemming from the belief that poor performance does not define their ability and instead offers valuable feedback for improvement. Findings from a study by Huang et al.²³ in 2022 suggest that students who are incremental theorists believe in the importance of effort and persistence in improving abilities, which ultimately enhances their sense of self-determination and intrinsic motivation to learn. This belief enables students to retain high academic performance even with the removal of extrinsic rewards such as academic awards, developing essential lifelong learning habits.¹⁰ Students who hold incremental mindsets may also be less likely to succumb to academic obstacles, instead looking for solutions to resolve barriers due to an intrinsic drive to learn.

Intrinsic motivation, therefore, is often positively associated with growth mindsets. It is defined as the inherent satisfaction derived from performing an activity without external incentives,¹⁰ completing tasks due to their own enjoyment and genuine interest.²⁴ It has become an important determinant in engagement in tasks that require persistence and high levels of efforts to achieve,²⁴ suggesting that intrinsic motivation often plays a key role in resilience when confronting barriers and obstacles faced throughout learning.

While incremental theory and intrinsic motivation are fundamentally different concepts, they are both factors that increase the inherent enjoyment of activities and the motivation to complete them. Although incremental theory and intrinsic motivation are incredibly beneficial tools for student performance, they are also important factors which contribute to lifelong learning and increase persistence when faced with challenges and setbacks.

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APPLICATION OF INCREMENTAL THEORY AND INTRINSIC MOTIVATION IN IMPROVING NON-ADHERENCE

While incremental theory and intrinsic motivation are historically rooted in educational settings, their core principles may help patients navigate around motivational barriers that arise from managing a chronic physiological condition like hypertension. As identified previously, motivational non-adherence for hypertensive patients stems from the lack of perceived value or need for completing their medication.¹⁷ This could be attributed to the oligosymptomatic nature of the disease, requiring long-lasting therapy that does not show immediate or satisfactory results, diminishing the perceived urgency of completing therapy schemes.¹³ It is also noted by Joho⁵ that an individual's perceived barrier to completing a medication course can be an important determinant in compliance. Hence, it is crucial to examine how incremental theory and intrinsic motivation could help alleviate these challenges.

As previously mentioned, intrinsic motivation plays a key role in regulating behaviours and satisfaction derived from completing a task. Intrinsically motivated individuals are driven to perform activities due to inherent enjoyment and high perceived value; therefore, external rewards are often not needed. In a medical setting, increasing the perceived value of taking medication could significantly improve patient adherence despite low-to-no external validation (i.e., symptom relief).

A study by Boaler²⁵ demonstrated individuals with a fixed mindset are less likely to persevere through challenges or respond critically to feedback due to the belief that capability and ability cannot be changed. In a healthcare setting, patients with fixed mindsets could be more likely to shy away from challenges or inconveniences that arise from adapting to the treatment regimen, for example, when management requires making lifestyle changes. Low perceived urgency and unsatisfactory results make compliance seem like an unnecessary burden, rather than recognising benefits in the long term.

However, by adopting a growth mindset, these challenges could potentially be mitigated. This is because individuals

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who possess implicit theories view challenges as an important step of their learning journey, as they believe that feedback received from evaluation opportunities can lead to improvement. This makes them more likely to confront and persist with solving obstacles compared to fixed mindset individuals, who believe that skill and talent are stagnant and cannot be altered with persistence and effort.²⁶ For a hypertensive patient, this may translate to viewing obstacles in their medication journey as a crucial opportunity for feedback response rather than an insurmountable roadblock. Alterations in lifestyle may be viewed as an opportunity to improve living conditions rather than a troublesome disruption in daily routines. Patients may also feel less intimidated by fear of potential challenges and be more willing to find alternative solutions instead of stopping treatment all together.

In essence, the lack of motivation to adhere to an asymptomatic treatment plan is fundamentally caused by low levels of perceived need to comply and shying away from potential obstacles. This leads to the conclusion that adoption of an incremental approach could intrinsically incentivise patients to complete therapy schemes despite differences between perceived benefit and actual benefit.

LIMITATIONS OF APPLICATIONS OF INCREMENTAL THEORIES

While growth mindset theory may be a possible solution to reducing motivational non-adherence, it is equally important to examine the potential limitations when it comes to reducing other barriers. As examined in Part 1, non-adherence is a multidimensional issue that arises from complex difficulties in three major areas: motivation, socioeconomic status and communication. Addressing these issues may involve constitutional changes in healthcare systems that require a large amount of effort and may take a long time.

For example, the majority of socioeconomic non-adherence stems from the inability to afford medication for long therapeutic time frames, often amplified when polypharmacy is involved. Addressing this issue may require reforming health policies, such as price regulation strategies²⁷ or providing healthcare coverage. Unemployment and poverty are also crucial determinants of non-adherence, as they are directly correlated with one's ability to afford medication for the duration of treatment. Reducing these factors may require systematic changes within government policies, such as providing benefits and pensions for those in need.

The majority of communicational non-adherence is caused by the inability to understand treatment plans, resulting in mistrust, misconception, and misunderstanding between provider and patient. This issue usually arises when there are language barriers that prevent effective communication, which requires systematic changes in the healthcare system such as employing more native-speaker doctors in areas that do not have English as an official language. Discriminatory behaviours enacted by providers can also greatly impact trust between patient-doctor relationships. This can be reduced by integrating anti-discriminatory classes into training programmes or fostering accountability for discriminatory practices.²⁸

Non-adherence is a complex issue that requires both individual and systematic changes, which may take long periods of time to implement and take effect. Although growth mindsets can be effectively used to reduce motivational challenges, it is also important to recognise that the observed phenomenon of non-adherence can reveal implicit flaws within healthcare systems.

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CONCLUSION

This literature demonstrates that the inherent link between intrinsic motivation and incremental theory holds significant application potential in medicine-related fields to reduce patient non-adherence, specifically by increasing both the internal drive to complete ongoing therapy plans without a direct external reward and the willingness to persevere despite obstacles encountered throughout treatment. However, external factors such as inability to afford long-term medication, lack of trust between provider and patient, and communication difficulties are also key contributors to non-adherence. These usually require constitutional changes within healthcare systems, such as making medication more affordable, resolving language barriers, and anti-discrimination training. For patient populations with sufficient functional literacy and the financial capability to afford sufficient medication, adopting an incremental mindset can be an effective strategy to manage motivation and routine-building. In comparison to other alternatives, fostering a growth mindset within patients can be more cost-effective, more affordable, and offer benefits outside of healthcare settings.

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